

Chapter 6. Power

6.1. Power Supplies

RP2350 requires five separate power supplies. However, in most applications, several of these can be combined and connected to a single power source. Typical applications only require a single 3.3 V supply. See [Figure 19](#).

The power supplies and a number of potential power supply schemes are described in the following sections. Detailed power supply parameters are provided in [Section 14.9.5](#).

6.1.1. Digital IO Supply (IOVDD)

IOVDD provides the IO supply for the chip's GPIO, and should be powered at a nominal voltage between 1.8 V and 3.3 V. The supply voltage sets the external signal level for the digital IO, and should be chosen based on the level required, see [Section 14.9](#) for details. All GPIOs share the same power supply and operate at the same signal level.

If the digital IO is powered at a nominal 1.8 V, the IO input thresholds should be adjusted by setting the [VOLTAGE_SELECT](#) register to 1. [VOLTAGE_SELECT](#) is set to 0 by default, which results in input thresholds that are valid for a nominal IO voltage between 2.5 V and 3.3 V. See [Chapter 9](#) for details.

CAUTION

Powering the IOVDD at 1.8 V with input thresholds set for a 2.5 V to 3.3 V supply is a safe operating mode, but will result in input thresholds that do not meet specification. Powering the IO at voltages greater than 1.8 V with input thresholds set for a 1.8 V supply may result in damage to the chip.

6.1.2. QSPI IO Supply (QSPI_IOVDD)

QSPI_IOVDD provides the IO supply for the chip's QSPI interface, and should be powered at a nominal voltage between 1.8 V and 3.3 V. The supply voltage sets the external signal level for the QSPI interface, and should be chosen based on the level required, see [Section 14.9](#) for details. In most applications the QSPI interface will be connected to an external flash device, which will determine the required signal level.

If the QSPI interface is powered at a nominal 1.8 V, the IO input thresholds should be adjusted by setting the [VOLTAGE_SELECT](#) register to 1. [VOLTAGE_SELECT](#) is set to 0 by default, which results in input thresholds that are valid for a nominal IO voltage between 2.5 V and 3.3 V. See [Chapter 9](#) for details.

CAUTION

Powering the IOVDD at 1.8 V with input thresholds set for a 2.5 V to 3.3 V supply is a safe operating mode, but will result in input thresholds that do not meet specification. Powering the IO at voltages greater than 1.8 V with input thresholds set for a 1.8 V supply may result in damage to the chip.

6.1.3. Digital Core Supply (DVDD)

The chip's core digital logic is powered by DVDD, which should be at a nominal 1.1 V. A dedicated on-chip core voltage regulator allows DVDD to be generated from a 2.7 V to 5.5 V input supply. See [Section 6.3](#) for details. Alternatively, DVDD can be supplied directly from an off-chip power source.

If the on-chip core voltage regulator is used, the two DVDD pins closest to the regulator should be decoupled with a 100nF capacitor close to the pins. The DVDD pin furthest from the regulator should be decoupled with a 4.7µF capacitor close to

the pin.

6.1.4. USB PHY and OTP Supply (USB_OTP_VDD)

USB_OTP_VDD supplies the chip's USB PHY and OTP memory, and should be powered at a nominal 3.3 V. To reduce the number of external power supplies, USB_OTP_VDD can use the same power source as the core voltage regulator analogue supply (VREG_AVDD), or digital IO supply (IOVDD), assuming IOVDD is also powered at 3.3 V. This supply must always be provided, even in applications where the USB PHY is never used.

USB_OTP_VDD should be decoupled with a 100nF capacitor close to the chip's USB_OTP_VDD pin.

6.1.5. ADC Supply (ADC_AVDD)

ADC_AVDD supplies the chip's Analogue to Digital Converter (ADC). It can be powered at a nominal voltage between 1.8 V and 3.3 V, but the performance of the ADC will be compromised at voltages below 2.97 V. To reduce the number of external power supplies, ADC_AVDD can use the same power source as the core voltage regulator analogue supply (VREG_AVDD) or digital IO supply (IOVDD).

i NOTE

It is safe to supply ADC_AVDD at a higher or lower voltage than IOVDD, e.g. to power the ADC at 3.3 V, for optimum performance, while supporting 1.8 V signal levels on the digital IO. But the voltage on the ADC analogue inputs must not exceed IOVDD, e.g. if IOVDD is powered at 1.8 V, the voltage on the ADC inputs should be limited to 1.8 V. Voltages greater than IOVDD will result in leakage currents through the ESD protection diodes. See [Section 14.9](#) for details.

ADC_AVDD should be decoupled with a 100nF capacitor close to the chip's ADC_AVDD pin.

6.1.6. Core Voltage Regulator Input Supply (VREG_VIN)

VREG_VIN is the input supply for the on-chip core voltage regulator, and should be in the range 2.7 V to 5.5 V. To reduce the number of external power supplies, VREG_VIN can use the same power source as the voltage regulator analogue supply (VREG_AVDD), or digital IO supply (IOVDD). Though care should be taken to minimise the noise on VREG_AVDD.

A 4.7µF capacitor should be connected between VREG_VIN and ground close to the chip's VREG_VIN pin.

For more details on the on-chip voltage regulator see [Section 6.3](#).

6.1.7. On-Chip Voltage Regulator Analogue Supply (VREG_AVDD)

VREG_AVDD supplies the on chip voltage regulator's analogue control circuits, and should be powered at a nominal 3.3 V. To reduce the number of external power supplies, VREG_AVDD can use the same power source as the voltage regulator input supply (VREG_VIN), or the digital IO supply (IOVDD). Though care should be taken to minimise the noise on VREG_AVDD. A passive low pass filter may be required, see [Section 6.3.7](#) for details.

i NOTE

VREG_AVDD also powers the chip's power-on reset and brownout detection blocks, so it must be powered even if the on-chip voltage regulator is not used.

6.1.8. Power Supply Sequencing

With the exception of the two voltage regulator supplies (**VREG_VIN** and **VREG_AVDD**), which should be powered up together, RP2350's power supplies may be powered up or down in any order. However, small transient currents may flow in the ADC supply (**ADC_AVDD**) if it is powered up before, or powered down after, the digital core supply (DVDD). This will not damage the chip, but can be avoided by powering up DVDD before or at the same time as **ADC_AVDD**, and powering down DVDD after or at the same time as **ADC_AVDD**. In the most common power supply scheme, where the chip is powered from a single 3.3 V supply, DVDD will be powered up shortly after **ADC_AVDD** due to the startup time of the on-chip voltage regulator. This is acceptable behaviour.

6.2. Power Management

RP2350 retains the power control features of RP2040, but extends them by splitting the chip's digital core into a number of power domains, which can be selectively powered off. This allows significant power saving in applications where the chip is not continuously active. This section describes the core power domains and how they are controlled. The legacy RP2040 power control features still offer useful power savings, and are described in [Section 6.5](#).

Power domains, and transitions between power states, are controlled by a Power manager. The Power manager runs from either an internal low power oscillator **lposc**, or the reference clock **clk_ref**. The device may be configured to power down under software control and can wakeup on a GPIO or timer event. Configuration of the power manager is via the **POWMAN** registers in [Section 6.4](#).

6.2.1. Core Power Domains

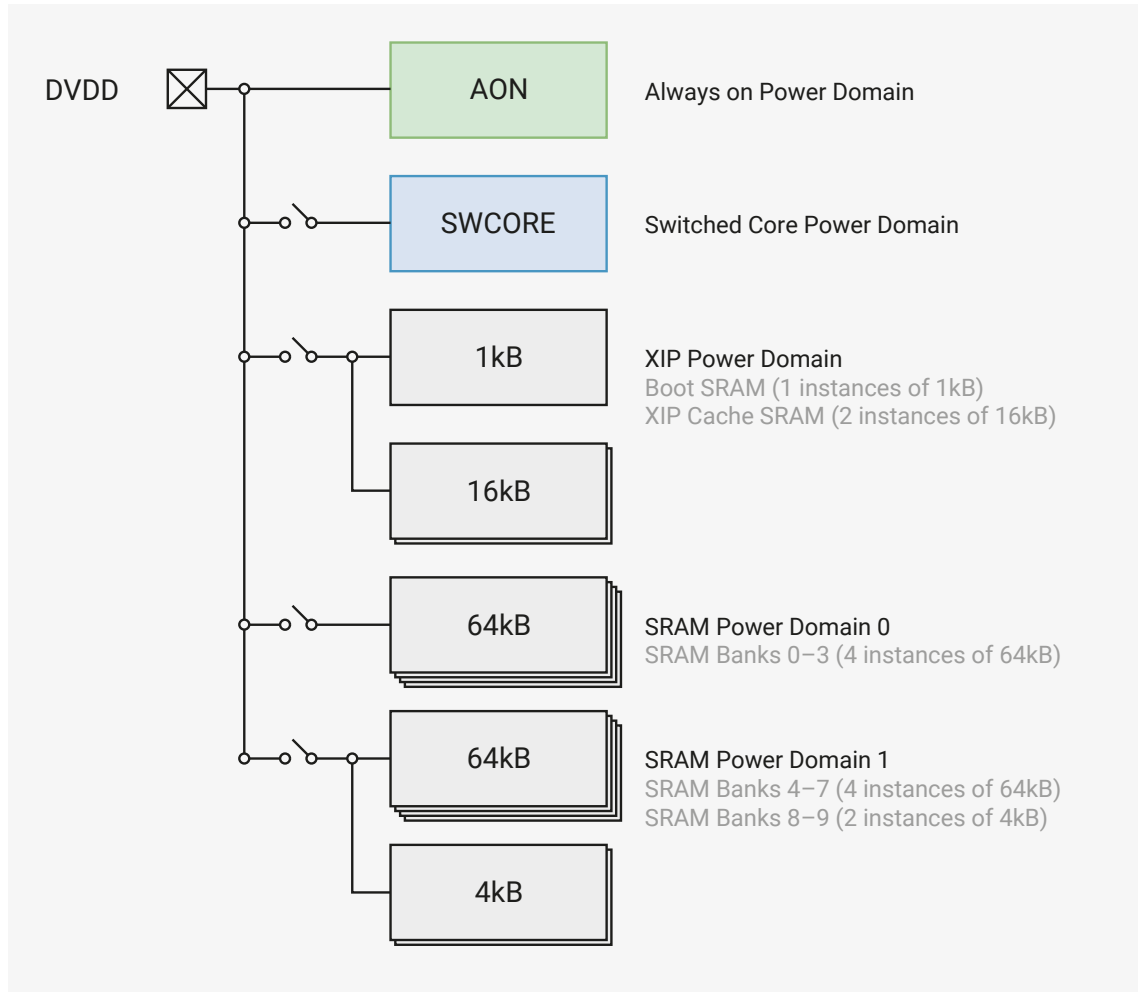
RP2350's core logic is divided into five power domains. With some restrictions, these domains can be selectively powered off to reduce the chip's power consumption. The five domains are:

- **AON** - Always On - a small amount of logic that is always powered on when chip's core supply (**DVDD**) is available
- **SWCORE** - Switched Core - the remaining core logic functions, including processors, bus fabric, peripherals, etc.
- **XIP** - XIP cache SRAM and Boot RAM
- **SRAM0** - SRAM Power Domain 0 - the lower half of the large SRAM banks
- **SRAM1** - SRAM Power Domain 1 - the upper half of the large SRAM banks, and the scratch SRAMs

Logic in the AON domain controls the power state of the other power domains, which can be powered on or off independently. The only exception is the XIP domain, which must always be powered when the SWCORE domain is powered. SRAMs that are powered on retain their contents when the switched core is powered off.

[Figure 18](#) gives an overview of the core power domains.

Figure 18. core power domains



6.2.2. Power States

RP2350 can operate in a number of power states, depending on which domains are powered on or off. Power states have names in the form **Pc.m** where:

- c** indicates the state of the switched core (SWCORE) domain: **0** = on / **1** = off
- m** is a 3 bit binary representation of the memory power domains, in the order XIP, SRAM0, SRAM1

P0.m states, where the switched core is powered on, are *Normal Operating* states. **P1.m** states, where the switched core is powered off, are *Low Power* states

Table 474 shows the available power states.

Table 474. supported power states

Power State	Description	AON	SWCORE	XIP	SRAM0	SRAM1
P0.0	Normal Operation	on	on	on	on	on
P0.1	Normal Operation (SRAM1 off)	on	on	on	on	off
P0.2	Normal Operation (SRAM0 off)	on	on	on	off	on
P0.3	Normal Operation (SRAM0 & SRAM1 off)	on	on	on	off	off
P1.0	Low Power	on	off	on	on	on
P1.1	Low Power (SRAM1 off)	on	off	on	on	off
P1.2	Low Power (SRAM0 off)	on	off	on	off	on

Power State	Description	AON	SWCORE	XIP	SRAM0	SRAM1
P1.3	Low Power (SRAM0 & SRAM1 off)	on	off	on	off	off
P1.4	Low Power (XIP off)	on	off	off	on	on
P1.5	Low Power (XIP & SRAM1 off)	on	off	off	on	off
P1.6	Low Power (XIP & SRAM0 off)	on	off	off	off	on
P1.7	Low Power (XIP & SRAM0 & SRAM1 off)	on	off	off	off	off
OFF	Not Powered	off	off	off	off	off

In the **OFF** state, the chip has no external power and all domains are unpowered. The chip moves from **OFF** to **P0.0** automatically as soon as external power is applied.

To determine the current power state, read the **STATE.CURRENT** field. **CURRENT** is a 4 bit field representing the power state of the switched core and memory power domains.

6.2.3. Power State Transitions

Transitions between power states can be initiated by software, hardware, or via the chip's debug subsystem. Once initiated, transitions are managed by autonomous power sequencers in the chip's AON power domain. The power sequencers can be configured, in a limited way, via the **SEQ_CFG** register. The sequencers can also be observed and controlled, again in a limited way, via the RP-AP registers in the chip's debug subsystem. These registers are described in [Section 3.5.10](#).

Valid power state transitions are as follows:

- all transitions from one **P0.m** state (switched core powered on) to another **P0.m** state (switched core powered on), if they increase or decrease the number of SRAM domains that are powered on
- all transitions from a **P0.m** state (switched core powered on) to a **P1.m** state (switched core powered off), except transitions that would result in a powered off SRAM domain becoming powered on
- all transitions from a **P1.m** state (switched core powered off) to a **P0.m** state (switched core powered on), except transitions that would result in a powered on SRAM domain becoming powered off

Transitions from one **P1.m** state (switched core powered off) to another **P1.m** state (switched core powered off) are not supported, and will be prevented by the hardware.

Valid transitions are shown in the table below.

Table 475. valid power state transitions

From	To											
P0.0		P0.1	P0.2	P0.3	P1.0	P1.1	P1.2	P1.3	P1.4	P1.5	P1.6	P1.7
P0.1	P0.0			P0.3		P1.1		P1.3		P1.5		P1.7
P0.2	P0.0			P0.3			P1.2	P1.3			P1.6	P1.7
P0.3	P0.0	P0.1	P0.2					P1.3				P1.7
P1.0	P0.0											
P1.1	P0.0	P0.1										
P1.2	P0.0		P0.2									
P1.3	P0.0	P0.1	P0.2	P0.3								
P1.4	P0.0											
P1.5	P0.0	P0.1										

From	To											
P1.6	P0.0		P0.2									
P1.7	P0.0	P0.1	P0.2	P0.3								

6.2.3.1. Transitions from Normal Operating (P0.m) States

Transitions from a Normal Operating (P0.m) state to either a Low Power (P1.m) state, or another Normal Operating (P0.m) state, are initiated by writing to the **STATE.REQ** field. **REQ** is a 4-bit field representing the requested power state of the switched core and memory power domains. The **STATE.WAITING** field will be set immediately, followed by the **STATE.CHANGING** field, once the actual state change starts. If a transition to a Low Power (P0.m) state is requested, **WAITING** will remain set until the processors have gone into a low power state (via **__wfi()**). In the **WAITING** state, writing to the **STATE.REQ** field can change or cancel the initial request. The requested state can not be changed once in the **CHANGING** state.

A request to move to an unsupported state, or a state that would result in an invalid transition, causes the **STATE.BAD_SW_REQ** field to be set.

If a hardware power up request is received while in the **WAITING** state, the transition requested via **STATE.REQ** will be halted and the power up request completed. The **STATE.PWRUP_WHILE_WAITING** and **STATE.REQ_IGNORED** fields will be set.

On writing to **STATE.REQ**:

- If there is a pending power up request, **STATE.REQ_IGNORED** is set and no further action is taken
- If the requested state is invalid, **STATE.BAD_SW_REQ** is set and no further action is taken
- If the switched core is being powered off, **STATE.WAITING** is set until both processors enter **__wfi()**. After which **STATE.CHANGING** will be set, but no processors will be powered up to read the flag at this time
 - If there is a power up request while in **STATE.WAITING**, **STATE.PWRUP_WHILE_WAITING** is set, which can also raise an interrupt to bring the processors out of **__wfi()**. No further action is taken
 - You can get out of the **WAITING** state by writing a new request to **STATE.REQ** before both processors have gone into **__wfi()**
- Any state request that isn't powering down the switched core, such as powering up or down SRAM domain 0 or 1 starts immediately. Software should wait until **STATE.CHANGING** has cleared to know the power down sequence. Once the **STATE.CHANGING** flag is cleared **STATE.CURRENT** is updated.
- If powering up, software should also wait for **STATE.CHANGING** to make sure everything is powered up before continuing. In practice this is handled by the RP2350 bootrom.

Invalid state transitions are:

- any combination of power up and power down requests
- any request which would result in power down of XIP/bootRAM and power up of swcore

If XIP, boot RAM, sram0, or sram1 remain powered while swcore is powered off, the sram will automatically switch to a low power state. Stored data will be retained.

Before transitioning to a switched-core power down state (P1.m), software needs to configure:

- the GPIO wakeup conditions if required
- the wakeup alarm if required
- the return state of the SRAM0 & SRAM1 domains

6.2.3.2. Transitions from Low Power (P1.m) States

Transitions from **P1.m** to **P0.m** states are initiated by GPIO events or the timer alarm.

There are up to 5 wakeup sources:

- up to 4 GPIO wakeups (level high/low or falling edge/rising edge)
- 1 alarm wakeup

GPIO wakeups are configured by the [PWRUP0-PWRUP3](#) registers. The wakeups are not enabled until the power sequencer completes the power down operation.

The alarm wakeup is configured by writing to the [ALARM_TIME_15T00-ALARM_TIME_63T048](#) registers. The alarm wakeup has a resolution of 1ms. Once set, the alarm wakeup is armed by writing a 1 to both [TIMER.PWRUP_ON_ALARM](#) and [TIMER.ALARM_ENAB](#). If the alarm fires during the power down sequence, a power up sequence will start when the power down sequence completes.

The [LAST_SWCORE_PWRUP](#) register indicates which event caused the most recent power up.

6.2.3.3. Debugger Initiated Power State Transitions

The debugger can be used to trigger a power up sequence via the [CSYSPWRUPREQ](#) output from the [SW-DP CTRL/STAT](#) register. This powers all domains (i.e. returns to state [P0.0](#)) and also inhibits any further software initiated power state transitions.

When [CSYSPWRUPREQ](#) is asserted, the power sequencer will:

- complete any power state transitions that are in progress
- return to power state [P0.0](#)
- assert [CSYSPWRUPACK](#) to signal completion to the debug host

If [CSYSPWRUPREQ](#) is de-asserted then software initiated power transitions will be able to resume. The user can detect when a software requested transition is ignored because of [CSYSPWRUPREQ](#) using the following hints:

- Getting a [STATE.REQ_IGNORED](#) after a write to [STATE.REQ](#)
- [CURRENT_PWRUP_REQ](#) will have bit 5 (coresight) set
- Either:
 - Get the debugger to de-assert [CSYSPWRUPREQ](#) or
 - Mask out [CSYSPWRUPREQ](#) by setting [DBG_PWRCFG.IGNORE](#)

i NOTE

[DBG_PWRCFG.IGNORE](#) is useful to test going to sleep with a debugger attached or ignoring [CSYSPWRUPREQ](#). A debugger will likely leave [CSYSPWRUPREQ](#) set when disconnecting. It would be impossible to go to sleep after this without [DBG_PWRCFG.IGNORE](#).

6.2.3.4. Power mode aware GPIO Control

The power manager sequencer is able to switch the state of two GPIO outputs on entry to and exit from a **P1.m** state, i.e. one where the switched core is powered down. This allows external devices to be power-aware. The GPIOs switch to indicate the low power state *after* the core is powered down and switch to indicate the high power state *before* the core is powered up. This ensures the high power state of the external components always overlaps the high power state of the core. The GPIOs are configured by the [EXT_CTRL0](#) and [EXT_CTRL1](#) registers.

6.2.3.5. Isolation

When powering down swcore, the pad control and data signals are latched and isolated from the IO logic. This avoids transitions on pads which could potentially corrupt external components. On swcore power up, the isolation is not released automatically. The user releases the isolation by clearing the ISO field of the pad control register (for example `GPIO0.ISO`) once the IO logic has been configured.

6.3. Core Voltage Regulator

RP2350 provides an on-chip voltage regulator for its digital core supply (`DVDD`). The regulator requires a 2.7 V to 5.5 V input supply (`VREG_VIN`), allowing DVDD to be generated directly from a single lithium ion cell, or a USB power supply. A separate, nominally 3.3 V, low noise supply (`VREG_AVDD`) is required for the regulator's analogue control circuits. The regulator supports both switching and linear modes of regulation, allowing efficient operation at both high and low loads.

To allow the chip to start up, the regulator is enabled by default, and will power-up as soon as its supplies are available. The regulator starts in switching mode, with a nominal 1.1 V output, but its operating mode and output voltage can be changed once the chip is out of reset. The output voltage can be set in the range 0.55 V to 3.30 V, and the regulator can supply up to 200mA.

Although intended for the chip's digital core supply (`DVDD`), the regulator can be used for other purposes if DVDD is powered directly from an external power supply.

6.3.1. Operating Modes

The regulator has the following three modes of operation.

6.3.1.1. Normal Mode

In Normal mode, the regulator operates in a switching mode, and can supply up to 200mA. Normal mode is used for `P0.x` power states, when the chip's switched core is powered on. The regulator must be in Normal Mode *before* the core supply current is allowed to exceed 1mA. The regulator starts up in Normal mode when its input supplies are first applied.

6.3.1.2. Low Power Mode

In Low Power mode, the regulator operates in a linear mode, and can only supply up to 1mA. Low Power mode can be used for `P1.x` power states, where the chip's switched core is powered off. The core supply current must be less than 1mA *before* the regulator is moved to Low Power mode. The regulator's output voltage is limited to 1.3 V in Low Power mode.

CAUTION

In Low Power mode, the output of the regulator is directly connected to `DVDD`. It is not possible to disconnect the regulator from `DVDD` in this mode. Do not put the regulator into Low Power mode if `DVDD` is being powered from an external supply.

6.3.1.3. High Impedance Mode

In High Impedance mode, the regulator is disabled, its power consumption is minimised, and its outputs are set to a high impedance state. This mode should only be used if the digital core supply (`DVDD`) is provided by an external

regulator. If the on-chip regulator is supplying DVDD, entering high impedance mode causes a reset event, returning the on-chip regulator to Normal mode.

6.3.2. Software Control

⚠ WARNING

Once enabled, software control *cannot be disabled*.

The regulator can be directly controlled by software, but must first be unlocked by writing a 1 to the **UNLOCK** field in the **VREG_CTRL** register. Once unlocked, the regulator can be controlled via the **VREG** register.

The regulator's operating mode defaults to Normal, at initial power up or after a reset event, but can be switched to High Impedance by writing a 1 to the **VREG** register's **HIZ** field. The regulator's output voltage can be set by writing to the register's **VSEL** field, see the **VREG** register description for details on available settings. To prevent accidental over-voltage, the output voltage is limited to 1.3 V unless the **DISABLE_VOLTAGE_LIMIT** field in the **VREG_CTRL** is set. The output voltage defaults to 1.1 V at initial power-on or after a reset event.

The **UPDATE_IN_PROGRESS** field in the **VREG** register is set while the regulator's operating mode or output voltage are being updated. When **UPDATE_IN_PROGRESS** is set, writes to the register are ignored.

It is not possible to place the regulator in Low Power mode under software control, as the load current will exceed 1mA when software is running.

⚠ CAUTION

The regulator's output voltage can be varied between 0.55 V and 3.3 V, but RP2350 may not operate reliably with its digital core supply (**DVDD**) at a voltage other than 1.1 V.

6.3.3. Power Manager Control

The regulator's operating mode and output voltage can also be controlled by the Power Manager. Power Manager control is typically used when the chip enters or exits a low power (**P1.x**) state, when software may not be running.

In addition to Normal and High Impedance modes, Power Manager control allows the regulator to be placed in Low Power mode. By default, the regulator switches to Low Power mode when entering a low power (**P1.x**) state, and returns to Normal mode when returning to a normal (**P0.x**) state.

The operating mode and output voltage in the low power state are set by the values in the **VREG_LP_ENTRY** register. And the operating mode and output voltage to be used when the chip has returned to a normal state are set by values in the **VREG_LP_EXIT** register. The registers contain an additional **MODE** field that allows Low Power mode to be selected.

The values in the registers must be written by software *before* requesting a transition to a low power state, as software will not be running during or after the transition. The actual transitions to and from the low power state are handled by the Power Manager. Once the chip has returned to a normal state, software can be run and the regulator controlled directly. The values in the **VREG** register reflect the regulator's current operating mode and output voltage once the chip has returned to a normal state.

CAUTION

Low Power mode should only be used when the regulator is providing the chip's digital core supply (**DVDD**), as the regulator's low power output is connected to **DVDD** on chip.

6.3.4. Status

To determine the status of the regulator, read the **VREG_STS** register, which contains two fields:

- **VOUT_OK** indicates whether the voltage regulator's output is being correctly regulated. At power-on, **VOUT_OK** remains low until the regulator has started up and the output voltage reaches the **VOUT_OK** assertion threshold (**VOUT_OK_{TH.ASSERT}**). It then remains high until the voltage drops below the **VOUT_OK** de-assertion threshold (**VOUT_OK_{TH.DEASSERT}**), remaining low until the output voltage is above the assertion threshold again. **VOUT_OK_{TH.ASSERT}** is nominally 90% of the selected output voltage, 0.99 V if the selected output voltage is 1.1 V, and **VOUT_OK_{TH.DEASSERT}** is nominally 87% of the selected output voltage, 0.957 V if the selected output voltage is 1.1 V. See [Section 14.9.6](#) for details.
- **STARTUP** is high when the regulator is starting up, and remains high until the regulator's operating mode or output voltage are changed, either by software or the Power Manager

NOTE

Adjusting the output voltage to a higher voltage will cause **VOUT_OK** to go low until the assertion threshold for the higher voltage is reached. **VOUT_OK** will also go low if the regulator is placed in High Impedance mode.

6.3.5. Current Limit

The voltage regulator includes a current limit to prevent the load current exceeding the maximum rated value. The output voltage will not be regulated and will drop below the selected value when the current limit is active. See [Section 14.9.6](#) for details.

6.3.6. Over Temperature Protection

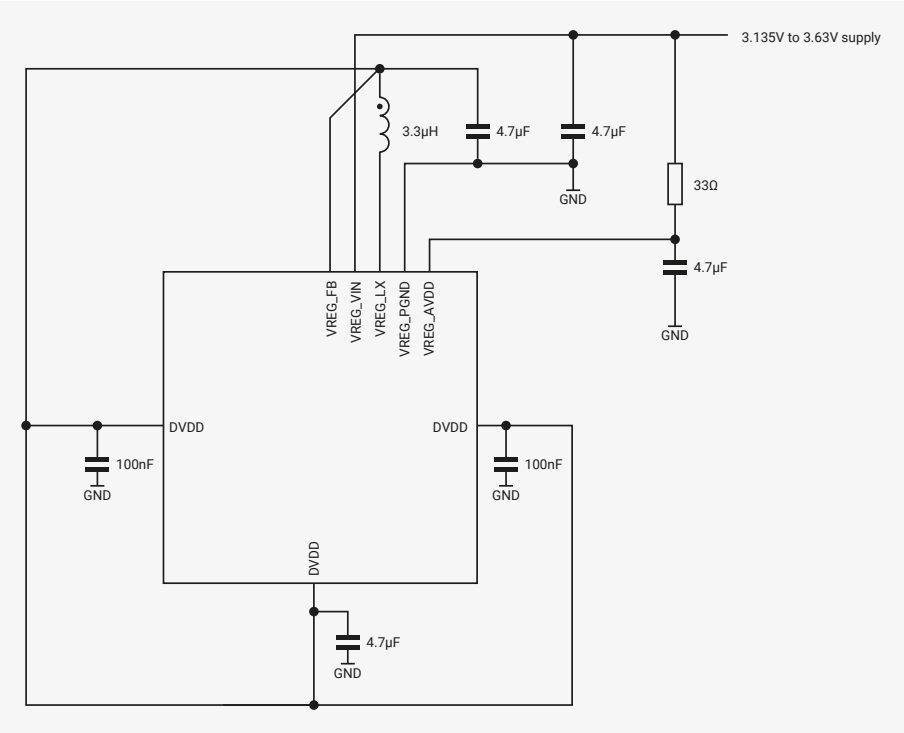
The voltage regulator will terminate regulation, and disable its power transistors, if the transistor junction temperature rises above a threshold set by the **HT_TH** field in the **VREG_CTRL** register. The regulator will restart regulation when the transistor junction temperature drops to approximately 20°C below the temperature threshold.

6.3.7. Application Circuit

The regulator requires two external power supplies, the input supply (**VREG_VIN**), and a separate low noise supply for its analogue control circuits (**VREG_AVDD**). **VREG_VIN** must be in the range 2.7 V to 5.5 V, and **VREG_AVDD** must be in the range 3.135 V to 3.63 V.

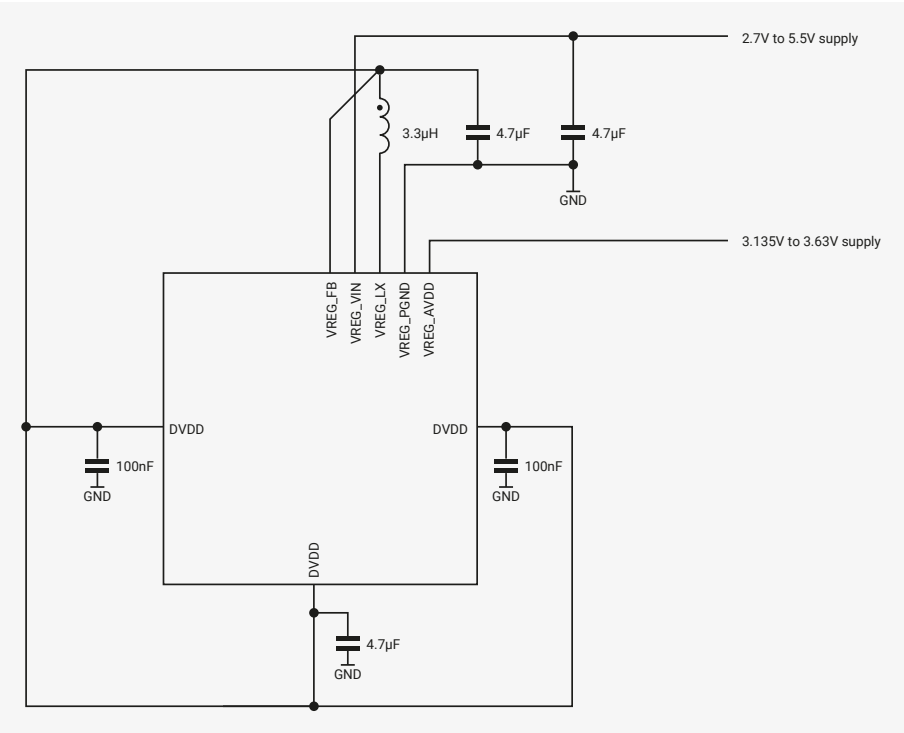
If **VREG_VIN** is limited to the range 3.135 V to 3.63 V, a single combined supply can be used for both **VREG_VIN** and **VREG_AVDD**. This approach is shown in [Figure 19](#). Care must be taken to minimise noise on **VREG_AVDD**.

Figure 19. core voltage regulator with combined supplies



Alternatively, to support input voltages above 3.63 V, VREG_VIN and VREG_AVDD can be powered separately. This is shown in [Figure 20](#).

Figure 20. core voltage regulator with separate supplies



6.3.8. External Components and PCB layout requirements

The most critical part of an RP2350 PCB layout is the core voltage regulator. This should be placed first on any board design and these guidelines must be strictly followed.

Figure 21. Regulator section of the Raspberry Pi Pico 2 schematic. The nets highlighted in bold show the high switching current paths

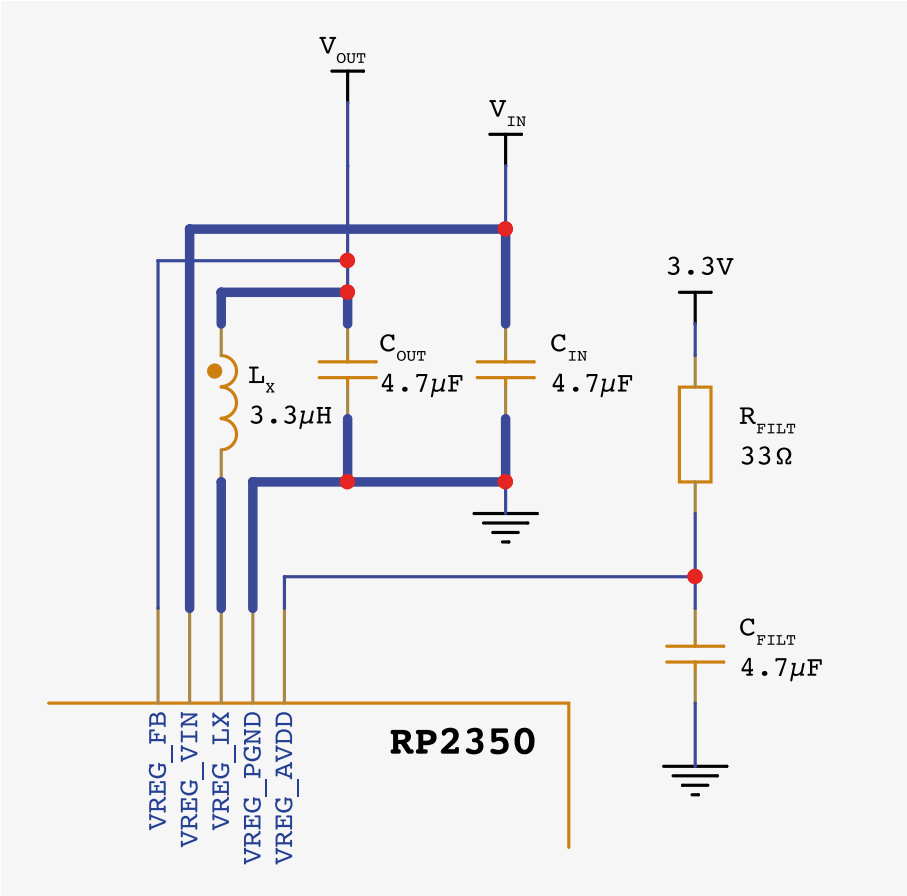
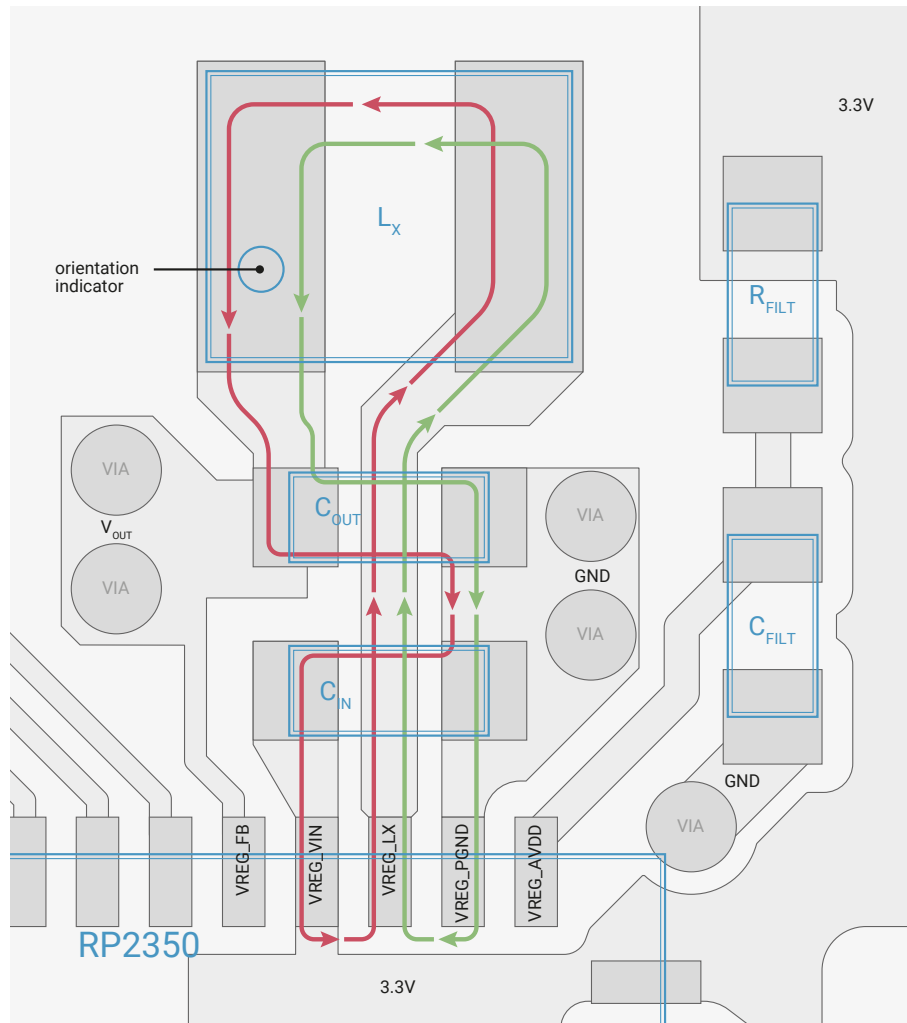


Figure 22. Regulator section of the Raspberry Pi Pico 2 PCB layout showing the high current paths for each of the regulator's switching phases. The AOTA-B201610S3R3-101-T inductor's case size is 0806 (2016 metric), the resistor and capacitors are 0402 (1005 metric)



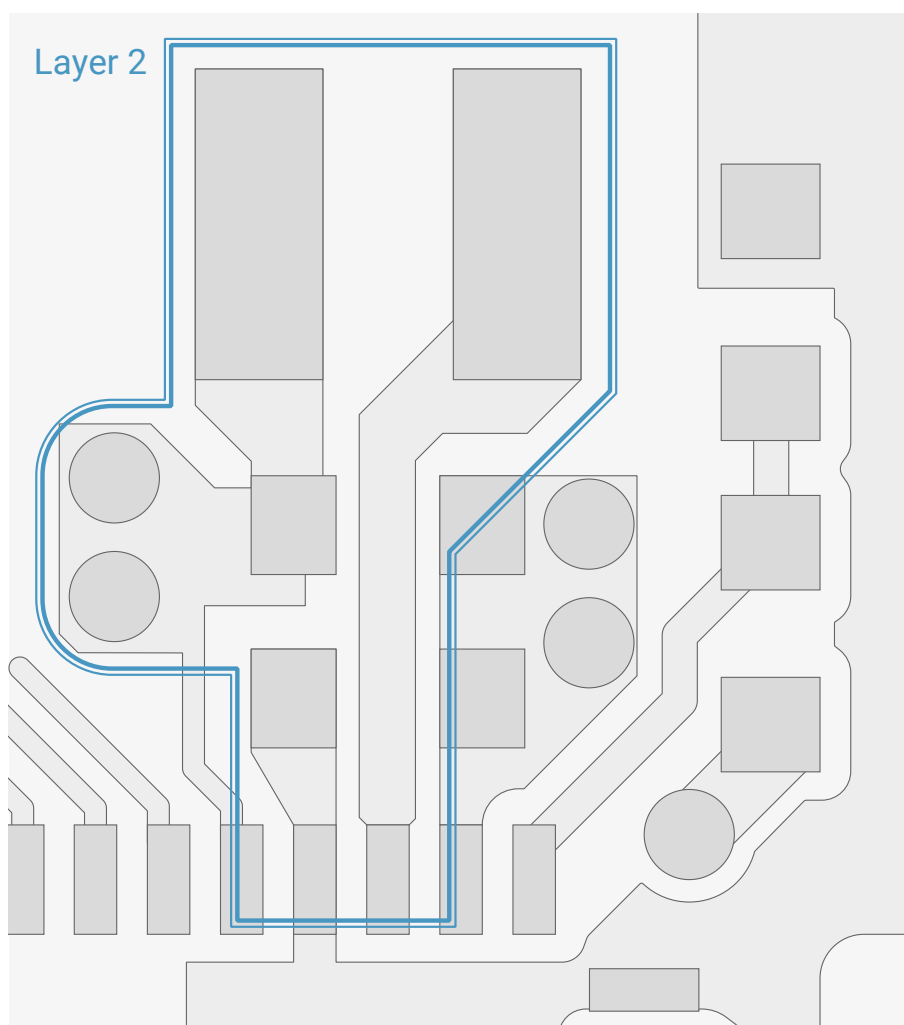
Designers should follow the above schematic [Figure 21](#) and layout [Figure 22](#) as closely as possible as this has had the most verification and is considered our best practice layout. This circuit design is present on the Raspberry Pi Pico 2 and RP2350 reference design (see [Hardware design with RP2350, Minimal Design Example](#)) and both of these designs are made available in either Cadence Allegro or Kicad formats respectively. [Figure 22](#) shows the regulator layout on the top layer of the Raspberry Pi Pico 2 PCB. The bottom layer under the regulator is a ground plane which connects to the QFN GND central pad.

6.3.8.1. Layout Recommendations

- **VREG_AVDD** is a noise sensitive signal and must be RC filtered as per [Figure 21](#).
 - Avoid doing anything that may couple noise into **VREG_AVDD**.
 - **C_{IN}** needs its own separate GND via / low impedance path back to the RP2350 GND pad.
- The red and green arrows in [Figure 22](#) show the high current paths for each of the regulator's switching phases. It is critical keep the loop area of these current paths as small and low-impedance as possible, while also keeping them isolated (i.e. only connect to main GND at one point).
 - Follow this layout as closely as possible.
 - Do not place any of **C_{IN}**/**L_X**/**C_{OUT}** on the opposite side of the PCB.
- Reduce parasitics on the **VREG_LX** node.
- On the top layer make sure to cut away any extra copper underneath the inductor, cut back copper near the **VREG_LX** trace where possible.

- For a multi-layer board (4 or more layers) please cut away any copper immediately underneath $L_X/VREG_LX$ node. For example, [Figure 23](#) illustrates this.
- The GND via placement is critical.
 - There must be a short-as-possible, low impedance GND path back to the Raspberry Pi Pico 2 QFN GND pad from the high-current GND at one single point (using 2 adjacent vias to reduce the impedance).
 - C_{FILT} must also have a low impedance and short-as-possible path back to the QFN GND pad (Do not share any GND vias with the C_{IN}/C_{OUT} high current GND).
- The VREG_FB pin should be fed from the output of C_{OUT} , avoiding routing directly underneath L_X .
- C_{OUT} is critical for regulator performance and EMI. It must be placed between $VREG_VIN$ and $VREG_PGND$ as close to the pins as practically possible.
 - In addition to C_{OUT} , for best performance we recommend a second $4.7\mu F$ capacitor is used on the V_{OUT} net, located on the bottom edge of the package (DVDD pin 23 on the QFN-60). Do not place this near L_X/C_{OUT} .

Figure 23. Cut-out beneath $L_X/VREG_LX$ net on layer 2 of 4 (or more) layer PCBs



6.3.8.2. Component Values

- C_{IN} should be at least $4.7\mu F$ and have a maximum parasitic resistance of $50m\Omega$.
- C_{OUT} must be $4.7\mu F \pm 20\%$ with a maximum parasitic resistance of $250m\Omega$ and a maximum inductance of $6nH$.
- L_X must be fully shielded, $3.3\mu H \pm 20\%$ and with a maximum DC resistance of $250m\Omega$. Saturation current should be at least $1.5A$. The inductor must be marked for polarity (see [Figure 24](#)) and placed on the layout as indicated in [Figure 22](#). As discussed below, we recommend the [AOTA-B201610S3R3-101-T](#).

6.3.8.3. Regulator Sensitivities

The RP2350 regulator has a few sensitivities:

- The **VREG_AVDD** supply is noise sensitive.
- Efficiency is quite sensitive to inductance roll-off with inductor current, so an inductor with low roll-off is required for best operation (generally the higher saturation current the better).
- Even with nominally fully shielded inductors, leakage magnetic field coupling into the loop formed by the output **VREG_LX** node through the inductor and output capacitor (C_{OUT}) seems to affect the regulator control loop and output voltage. Field orientation (and hence inductor orientation) matters - the inductor has to be the right way around to make sure the regulator operates properly especially at higher output currents and for higher load transients. This necessitates an inductor with marked polarity.

To meet the above requirements, Raspberry Pi have worked with Abracon to create a custom 2.0×1.6mm 3.3μH polarity-marked inductor, part number **AOTA-B201610S3R3-101-T** (see [Figure 24](#) and [Figure 24](#)). These will be available in general distribution in time, but for now please contact Raspberry Pi to request samples / production volumes.

Raspberry Pi is still working with the regulator IP vendor to fully verify and qualify the regulator and custom inductor.

Figure 24. AOTA-B201610S3R3-101-T inductor with orientation marking, showing current and magnetic field directions

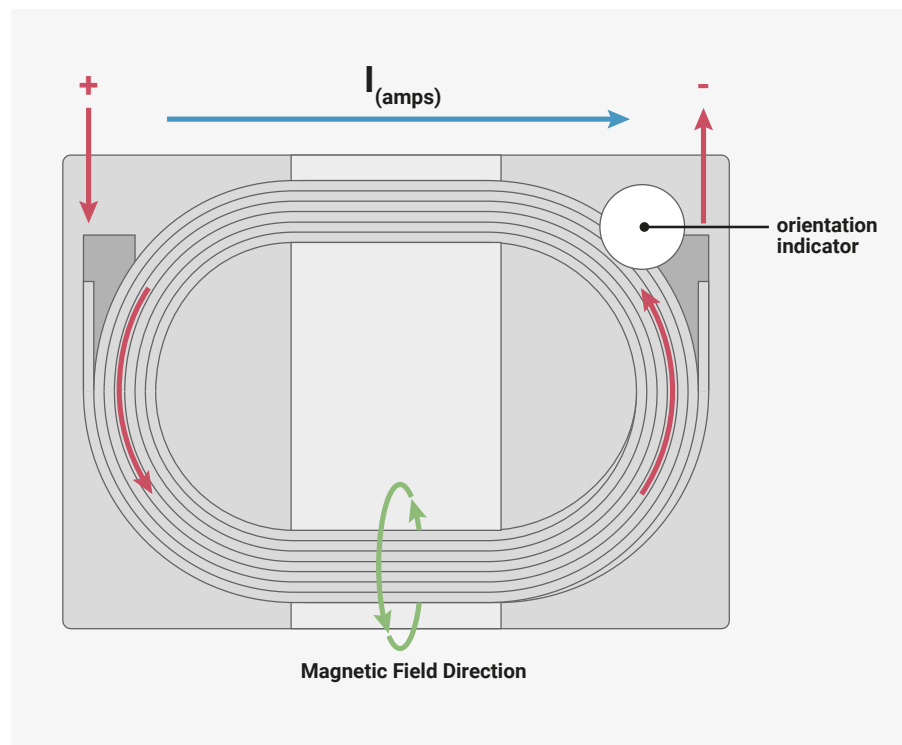
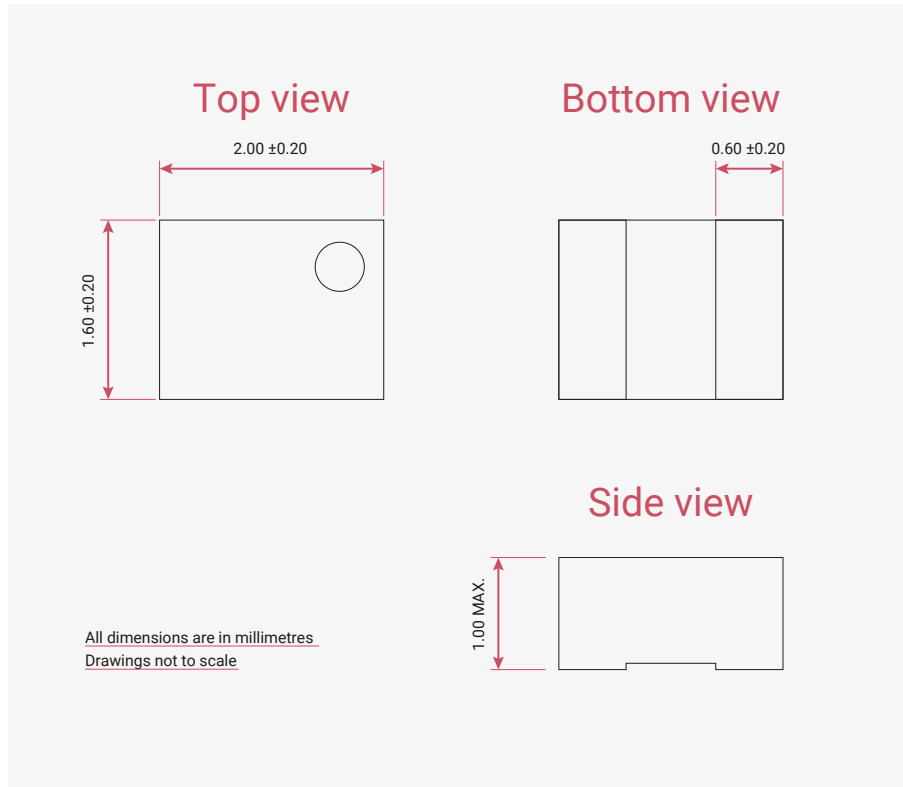


Figure 25. Dimensions of the AOTA-B201610S3R3-101-T inductor



6.3.9. List of Registers

The voltage regulator shares a register address space with other power management subsystems in the always-on domain. This address space is referred to as **POWMAN** elsewhere in this document, and a complete list of **POWMAN** registers is provided in [Section 6.4](#). For reference information on **POWMAN** registers associated with the voltage regulator is repeated here.

The **POWMAN** registers start at a base address of `0x40100000` (defined as **POWMAN_BASE** in the SDK).

- **VREG_CTRL**
- **VREG_STS**
- **VREG**
- **VREG_LP_ENTRY**
- **VREG_LP_EXIT**

6.4. Power Management (POWMAN) Registers

Password-protected **POWMAN** registers require a password (`0x5AFE`) to be written to the top 16 bits to enable the write operation. This protects against accidental writes which could crash the chip untraceably. Writes to protected registers that do not include the password are ignored, setting a flag in the **BADPASSWD** register. Reads from protected registers do not return the password, to protect against erroneous read-modify-write operations.

Protected registers obviously do not have writeable fields in the top 16 bits, however they may have read-only fields in that range.

All registers with address offsets up to and including `0x000000ac` are password protected. Therefore, the following writeable registers are unprotected and have 32-bit write access:

- POWMAN_SCRATCH0 → POWMAN_SCRATCH7
- POWMAN_BOOT0 → POWMAN_BOOT3
- POWMAN_INTR
- POWMAN_INTE
- POWMAN_INTF

Table 476. List of POWMAN registers

Offset	Name	Info
0x00	BADPASSWD	Indicates a bad password has been used
0x04	VREG_CTRL	Voltage Regulator Control
0x08	VREG_STS	Voltage Regulator Status
0x0c	VREG	Voltage Regulator Settings
0x10	VREG_LP_ENTRY	Voltage Regulator Low Power Entry Settings
0x14	VREG_LP_EXIT	Voltage Regulator Low Power Exit Settings
0x18	BOD_CTRL	Brown-out Detection Control
0x1c	BOD	Brown-out Detection Settings
0x20	BOD_LP_ENTRY	Brown-out Detection Low Power Entry Settings
0x24	BOD_LP_EXIT	Brown-out Detection Low Power Exit Settings
0x28	LPOSC	Low power oscillator control register.
0x2c	CHIP_RESET	Chip reset control and status
0x30	WDSEL	Allows a watchdog reset to reset the internal state of powman in addition to the power-on state machine (PSM). Note that powman ignores watchdog resets that do not select at least the CLOCKS stage or earlier stages in the PSM. If using these bits, it's recommended to set PSM_WDSEL to all-ones in addition to the desired bits in this register. Failing to select CLOCKS or earlier will result in the POWMAN_WDSEL register having no effect.
0x34	SEQ_CFG	For configuration of the power sequencer Writes are ignored while POWMAN_STATE_CHANGING=1

Offset	Name	Info
0x38	STATE	This register controls the power state of the 4 power domains. The current power state is indicated in POWMAN_STATE_CURRENT which is read-only. To change the state, write to POWMAN_STATE_REQ. The coding of POWMAN_STATE_CURRENT & POWMAN_STATE_REQ corresponds to the power states defined in the datasheet: - bit 3 = SWCORE - bit 2 = XIP cache - bit 1 = SRAM0 - bit 0 = SRAM1 0 = powered up 1 = powered down When POWMAN_STATE_REQ is written, the POWMAN_STATE_WAITING flag is set while the Power Manager determines what is required. If an invalid transition is requested the Power Manager will still register the request in POWMAN_STATE_REQ but will also set the POWMAN_BAD_REQ flag. It will then implement the power-up requests and ignore the power down requests. To do nothing would risk entering an unrecoverable lock-up state. Invalid requests are: any combination of power up and power down requests any request that results in swcore being powered and xip unpowered If the request is to power down the switched-core domain then POWMAN_STATE_WAITING stays active until the processors halt. During this time the POWMAN_STATE_REQ field can be re-written to change or cancel the request. When the power state transition begins the POWMAN_STATE_WAITING flag is cleared, the POWMAN_STATE_CHANGING flag is set and POWMAN register writes are ignored until the transition completes.
0x3c	POW_FASTDIV	
0x40	POW_DELAY	power state machine delays
0x44	EXT_CTRL0	Configures a gpio as a power mode aware control output
0x48	EXT_CTRL1	Configures a gpio as a power mode aware control output
0x4c	EXT_TIME_REF	Select a GPIO to use as a time reference, the source can be used to drive the low power clock at 32kHz, or to provide a 1ms tick to the timer, or provide a 1Hz tick to the timer. The tick selection is controlled by the POWMAN_TIMER register.
0x50	LPOSC_FREQ_KHZ_INT	Informs the AON Timer of the integer component of the clock frequency when running off the LPOSC.
0x54	LPOSC_FREQ_KHZ_FRAC	Informs the AON Timer of the fractional component of the clock frequency when running off the LPOSC.
0x58	XOSC_FREQ_KHZ_INT	Informs the AON Timer of the integer component of the clock frequency when running off the XOSC.
0x5c	XOSC_FREQ_KHZ_FRAC	Informs the AON Timer of the fractional component of the clock frequency when running off the XOSC.
0x60	SET_TIME_63T048	
0x64	SET_TIME_47T032	

Offset	Name	Info
0x68	SET_TIME_31T016	
0x6c	SET_TIME_15T00	
0x70	READ_TIME_UPPER	
0x74	READ_TIME_LOWER	
0x78	ALARM_TIME_63T048	
0x7c	ALARM_TIME_47T032	
0x80	ALARM_TIME_31T016	
0x84	ALARM_TIME_15T00	
0x88	TIMER	
0x8c	PWRUP0	4 GPIO powerup events can be configured to wake the chip up from a low power state. The pwrups are level/edge sensitive and can be set to trigger on a high/rising or low/falling event The number of gpios available depends on the package option. An invalid selection will be ignored source = 0 selects gpio0 . . source = 47 selects gpio47 source = 48 selects qspi_ss source = 49 selects qspi_sd0 source = 50 selects qspi_sd1 source = 51 selects qspi_sd2 source = 52 selects qspi_sd3 source = 53 selects qspi_sclk level = 0 triggers the pwrup when the source is low level = 1 triggers the pwrup when the source is high
0x90	PWRUP1	4 GPIO powerup events can be configured to wake the chip up from a low power state. The pwrups are level/edge sensitive and can be set to trigger on a high/rising or low/falling event The number of gpios available depends on the package option. An invalid selection will be ignored source = 0 selects gpio0 . . source = 47 selects gpio47 source = 48 selects qspi_ss source = 49 selects qspi_sd0 source = 50 selects qspi_sd1 source = 51 selects qspi_sd2 source = 52 selects qspi_sd3 source = 53 selects qspi_sclk level = 0 triggers the pwrup when the source is low level = 1 triggers the pwrup when the source is high

Offset	Name	Info
0x94	PWRUP2	4 GPIO powerup events can be configured to wake the chip up from a low power state. The pwrups are level/edge sensitive and can be set to trigger on a high/rising or low/falling event The number of gpios available depends on the package option. An invalid selection will be ignored source = 0 selects gpio0 . . source = 47 selects gpio47 source = 48 selects qspi_ss source = 49 selects qspi_sd0 source = 50 selects qspi_sd1 source = 51 selects qspi_sd2 source = 52 selects qspi_sd3 source = 53 selects qspi_sclk level = 0 triggers the pwrup when the source is low level = 1 triggers the pwrup when the source is high
0x98	PWRUP3	4 GPIO powerup events can be configured to wake the chip up from a low power state. The pwrups are level/edge sensitive and can be set to trigger on a high/rising or low/falling event The number of gpios available depends on the package option. An invalid selection will be ignored source = 0 selects gpio0 . . source = 47 selects gpio47 source = 48 selects qspi_ss source = 49 selects qspi_sd0 source = 50 selects qspi_sd1 source = 51 selects qspi_sd2 source = 52 selects qspi_sd3 source = 53 selects qspi_sclk level = 0 triggers the pwrup when the source is low level = 1 triggers the pwrup when the source is high
0x9c	CURRENT_PWRUP_REQ	Indicates current powerup request state pwrup events can be cleared by removing the enable from the pwrup register. The alarm pwrup req can be cleared by clearing timer.alarm_enab 0 = chip reset, for the source of the last reset see POWMAN_CHIP_RESET 1 = pwrup0 2 = pwrup1 3 = pwrup2 4 = pwrup3 5 = coresize_pwrup 6 = alarm_pwrup

Offset	Name	Info
0xa0	LAST_SWCORE_PWRUP	Indicates which pwrup source triggered the last switched-core power up 0 = chip reset, for the source of the last reset see POWMAN_CHIP_RESET 1 = pwrup0 2 = pwrup1 3 = pwrup2 4 = pwrup3 5 = coresight_pwrup 6 = alarm_pwrup
0xa4	DBG_PWRCFG	
0xa8	BOOTDIS	Tell the bootrom to ignore the BOOT0..3 registers following the next RSM reset (e.g. the next core power down/up). If an early boot stage has soft-locked some OTP pages in order to protect their contents from later stages, there is a risk that Secure code running at a later stage can unlock the pages by powering the core up and down. This register can be used to ensure that the bootloader runs as normal on the next power up, preventing Secure code at a later stage from accessing OTP in its unlocked state. Should be used in conjunction with the OTP BOOTDIS register.
0xac	DBGCONFIG	
0xb0	SCRATCH0	Scratch register. Information persists in low power mode
0xb4	SCRATCH1	Scratch register. Information persists in low power mode
0xb8	SCRATCH2	Scratch register. Information persists in low power mode
0xbc	SCRATCH3	Scratch register. Information persists in low power mode
0xc0	SCRATCH4	Scratch register. Information persists in low power mode
0xc4	SCRATCH5	Scratch register. Information persists in low power mode
0xc8	SCRATCH6	Scratch register. Information persists in low power mode
0xcc	SCRATCH7	Scratch register. Information persists in low power mode
0xd0	BOOT0	Scratch register. Information persists in low power mode
0xd4	BOOT1	Scratch register. Information persists in low power mode
0xd8	BOOT2	Scratch register. Information persists in low power mode
0xdc	BOOT3	Scratch register. Information persists in low power mode
0xe0	INTR	Raw Interrupts
0xe4	INTE	Interrupt Enable
0xe8	INTF	Interrupt Force
0xec	INTS	Interrupt status after masking & forcing

POWMAN: BADPASSWD Register

Offset: 0x00

Table 477.
BADPASSWD Register

Bits	Description	Type	Reset
31:1	Reserved.	-	-
0	Indicates a bad password has been used	WC	0x0

POWMAN: VREG_CTRL Register

Offset: 0x04

Description

Voltage Regulator Control

Table 478.
VREG_CTRL Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15	RST_N : returns the regulator to its startup settings 0 - reset 1 - not reset (default)	RW	0x1
14	Reserved.	-	-
13	UNLOCK : unlocks the VREG control interface after power up 0 - Locked (default) 1 - Unlocked It cannot be relocked when it is unlocked.	RW	0x0
12	ISOLATE : isolates the VREG control interface 0 - not isolated (default) 1 - isolated	RW	0x0
11:9	Reserved.	-	-
8	DISABLE_VOLTAGE_LIMIT : 0=not disabled, 1=enabled	RW	0x0
7	Reserved.	-	-
6:4	HT_TH : high temperature protection threshold regulator power transistors are disabled when junction temperature exceeds threshold 000 - 100C 001 - 105C 010 - 110C 011 - 115C 100 - 120C 101 - 125C 110 - 135C 111 - 150C	RW	0x5
3:2	Reserved.	-	-
1:0	RESERVED : write 0 to this field	RW	0x0

POWMAN: VREG_STS Register

Offset: 0x08

Description

Voltage Regulator Status

Table 479. VREG_STS
Register

Bits	Description	Type	Reset
31:5	Reserved.	-	-
4	VOUT_OK : output regulation status 0=not in regulation, 1=in regulation	RO	0x0
3:1	Reserved.	-	-
0	STARTUP : startup status 0=startup complete, 1=starting up	RO	0x0

POWMAN: VREG Register

Offset: 0x0c

Description

Voltage Regulator Settings

Table 480. VREG Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15	UPDATE_IN_PROGRESS : regulator state is being updated writes to the vreg register will be ignored when this field is set	RO	0x0
14:9	Reserved.	-	-

Bits	Description	Type	Reset
8:4	VSEL : output voltage select the regulator output voltage is limited to 1.3V unless the voltage limit is disabled using the disable_voltage_limit field in the vreg_ctrl register 00000 - 0.55V 00001 - 0.60V 00010 - 0.65V 00011 - 0.70V 00100 - 0.75V 00101 - 0.80V 00110 - 0.85V 00111 - 0.90V 01000 - 0.95V 01001 - 1.00V 01010 - 1.05V 01011 - 1.10V (default) 01100 - 1.15V 01101 - 1.20V 01110 - 1.25V 01111 - 1.30V 10000 - 1.35V 10001 - 1.40V 10010 - 1.50V 10011 - 1.60V 10100 - 1.65V 10101 - 1.70V 10110 - 1.80V 10111 - 1.90V 11000 - 2.00V 11001 - 2.35V 11010 - 2.50V 11011 - 2.65V 11100 - 2.80V 11101 - 3.00V 11110 - 3.15V 11111 - 3.30V	RW	0x0b
3	Reserved.	-	-
2	RESERVED : write 0 to this field	RW	0x0
1	HIZ : high impedance mode select 0=not in high impedance mode, 1=in high impedance mode	RW	0x0
0	Reserved.	-	-

POWMAN: VREG_LP_ENTRY Register

Offset: 0x10

Description

Voltage Regulator Low Power Entry Settings

Table 481.
VREG_LP_ENTRY
Register

Bits	Description	Type	Reset
31:9	Reserved.	-	-

Bits	Description	Type	Reset
8:4	VSEL : output voltage select the regulator output voltage is limited to 1.3V unless the voltage limit is disabled using the disable_voltage_limit field in the vreg_ctrl register 00000 - 0.55V 00001 - 0.60V 00010 - 0.65V 00011 - 0.70V 00100 - 0.75V 00101 - 0.80V 00110 - 0.85V 00111 - 0.90V 01000 - 0.95V 01001 - 1.00V 01010 - 1.05V 01011 - 1.10V (default) 01100 - 1.15V 01101 - 1.20V 01110 - 1.25V 01111 - 1.30V 10000 - 1.35V 10001 - 1.40V 10010 - 1.50V 10011 - 1.60V 10100 - 1.65V 10101 - 1.70V 10110 - 1.80V 10111 - 1.90V 11000 - 2.00V 11001 - 2.35V 11010 - 2.50V 11011 - 2.65V 11100 - 2.80V 11101 - 3.00V 11110 - 3.15V 11111 - 3.30V	RW	0x0b
3	Reserved.	-	-
2	MODE : selects either normal (switching) mode or low power (linear) mode low power mode can only be selected for output voltages up to 1.3V 0 = normal mode (switching) 1 = low power mode (linear)	RW	0x1
1	HIZ : high impedance mode select 0=not in high impedance mode, 1=in high impedance mode	RW	0x0
0	Reserved.	-	-

POWMAN: VREG_LP_EXIT Register

Offset: 0x14

Description

Voltage Regulator Low Power Exit Settings

Table 482.
VREG_LP_EXIT
Register

Bits	Description	Type	Reset
31:9	Reserved.	-	-
8:4	VSEL : output voltage select the regulator output voltage is limited to 1.3V unless the voltage limit is disabled using the disable_voltage_limit field in the vreg_ctrl register 00000 - 0.55V 00001 - 0.60V 00010 - 0.65V 00011 - 0.70V 00100 - 0.75V 00101 - 0.80V 00110 - 0.85V 00111 - 0.90V 01000 - 0.95V 01001 - 1.00V 01010 - 1.05V 01011 - 1.10V (default) 01100 - 1.15V 01101 - 1.20V 01110 - 1.25V 01111 - 1.30V 10000 - 1.35V 10001 - 1.40V 10010 - 1.50V 10011 - 1.60V 10100 - 1.65V 10101 - 1.70V 10110 - 1.80V 10111 - 1.90V 11000 - 2.00V 11001 - 2.35V 11010 - 2.50V 11011 - 2.65V 11100 - 2.80V 11101 - 3.00V 11110 - 3.15V 11111 - 3.30V	RW	0x0b
3	Reserved.	-	-
2	MODE : selects either normal (switching) mode or low power (linear) mode low power mode can only be selected for output voltages up to 1.3V 0 = normal mode (switching) 1 = low power mode (linear)	RW	0x0
1	HIZ : high impedance mode select 0=not in high impedance mode, 1=in high impedance mode	RW	0x0
0	Reserved.	-	-

POWMAN: BOD_CTRL Register

Offset: 0x18

Description

Brown-out Detection Control

Table 483. BOD_CTRL Register

Bits	Description	Type	Reset
31:13	Reserved.	-	-
12	ISOLATE : isolates the brown-out detection control interface 0 - not isolated (default) 1 - isolated	RW	0x0
11:0	Reserved.	-	-

POWMAN: BOD Register

Offset: 0x1c

Description

Brown-out Detection Settings

Table 484. BOD Register

Bits	Description	Type	Reset
31:9	Reserved.	-	-
8:4	VSEL : threshold select 00000 - 0.473V 00001 - 0.516V 00010 - 0.559V 00011 - 0.602V 00100 - 0.645VS 00101 - 0.688V 00110 - 0.731V 00111 - 0.774V 01000 - 0.817V 01001 - 0.860V (default) 01010 - 0.903V 01011 - 0.946V 01100 - 0.989V 01101 - 1.032V 01110 - 1.075V 01111 - 1.118V 10000 - 1.161V 10001 - 1.204V	RW	0x0b
3:1	Reserved.	-	-
0	EN : enable brown-out detection 0=not enabled, 1=enabled	RW	0x1

POWMAN: BOD_LP_ENTRY Register

Offset: 0x20

Description

Brown-out Detection Low Power Entry Settings

Table 485. BOD_LP_ENTRY Register

Bits	Description	Type	Reset
31:9	Reserved.	-	-

Bits	Description	Type	Reset
8:4	VSEL : threshold select 00000 - 0.473V 00001 - 0.516V 00010 - 0.559V 00011 - 0.602V 00100 - 0.645VS 00101 - 0.688V 00110 - 0.731V 00111 - 0.774V 01000 - 0.817V 01001 - 0.860V (default) 01010 - 0.903V 01011 - 0.946V 01100 - 0.989V 01101 - 1.032V 01110 - 1.075V 01111 - 1.118V 10000 - 1.161V 10001 - 1.204V	RW	0x0b
3:1	Reserved.	-	-
0	EN : enable brown-out detection 0=not enabled, 1=enabled	RW	0x0

POWMAN: BOD_LP_EXIT Register

Offset: 0x24

Description

Brown-out Detection Low Power Exit Settings

Table 486.
BOD_LP_EXIT Register

Bits	Description	Type	Reset
31:9	Reserved.	-	-
8:4	VSEL : threshold select 00000 - 0.473V 00001 - 0.516V 00010 - 0.559V 00011 - 0.602V 00100 - 0.645VS 00101 - 0.688V 00110 - 0.731V 00111 - 0.774V 01000 - 0.817V 01001 - 0.860V (default) 01010 - 0.903V 01011 - 0.946V 01100 - 0.989V 01101 - 1.032V 01110 - 1.075V 01111 - 1.118V 10000 - 1.161V 10001 - 1.204V	RW	0x0b
3:1	Reserved.	-	-

Bits	Description	Type	Reset
0	EN : enable brown-out detection 0=not enabled, 1=enabled	RW	0x1

POWMAN: LPOSC Register

Offset: 0x28

Description

Low power oscillator control register.

Table 487. LPOSC Register

Bits	Description	Type	Reset
31:10	Reserved.	-	-
9:4	TRIM : Frequency trim - the trim step is typically 1% of the reset frequency, but can be up to 3%	RW	0x20
3:2	Reserved.	-	-
1:0	MODE : This feature has been removed	RW	0x3

POWMAN: CHIP_RESET Register

Offset: 0x2c

Description

Chip reset control and status

Table 488. CHIP_RESET Register

Bits	Description	Type	Reset
31:29	Reserved.	-	-
28	HAD_WATCHDOG_RESET_PSM : Last reset was a watchdog timeout which was configured to reset the power-on state machine This resets: double_tap flag no DP no RPAP no rescue_flag no timer no powman no swcore no psm yes and does not change the power state	RO	0x0
27	HAD_HZD_SYS_RESET_REQ : Last reset was a system reset from the hazard debugger This resets: double_tap flag no DP no RPAP no rescue_flag no timer no powman no swcore no psm yes and does not change the power state	RO	0x0

Bits	Description	Type	Reset
26	HAD_GLITCH_DETECT: Last reset was due to a power supply glitch This resets: double_tap flag no DP no RPAP no rescue_flag no timer no powman no swcore no psm yes and does not change the power state	RO	0x0
25	HAD_SWCORE_PD: Last reset was a switched core powerdown This resets: double_tap flag no DP no RPAP no rescue_flag no timer no powman no swcore yes psm yes then starts the power sequencer	RO	0x0
24	HAD_WATCHDOG_RESET_SWCORE: Last reset was a watchdog timeout which was configured to reset the switched-core This resets: double_tap flag no DP no RPAP no rescue_flag no timer no powman no swcore yes psm yes then starts the power sequencer	RO	0x0
23	HAD_WATCHDOG_RESET_POWMAN: Last reset was a watchdog timeout which was configured to reset the power manager This resets: double_tap flag no DP no RPAP no rescue_flag no timer yes powman yes swcore yes psm yes then starts the power sequencer	RO	0x0

Bits	Description	Type	Reset
22	HAD_WATCHDOG_RESET_POWMAN_ASYNC: Last reset was a watchdog timeout which was configured to reset the power manager asynchronously This resets: double_tap flag no DP no RPAP no rescue_flag no timer yes powman yes swcore yes psm yes then starts the power sequencer	RO	0x0
21	HAD_RESCUE: Last reset was a rescue reset from the debugger This resets: double_tap flag no DP no RPAP no rescue_flag no, it sets this flag timer yes powman yes swcore yes psm yes then starts the power sequencer	RO	0x0
20	Reserved.	-	-
19	HAD_DP_RESET_REQ: Last reset was an reset request from the arm debugger This resets: double_tap flag no DP no RPAP no rescue_flag yes timer yes powman yes swcore yes psm yes then starts the power sequencer	RO	0x0
18	HAD_RUN_LOW: Last reset was from the RUN pin This resets: double_tap flag no DP yes RPAP yes rescue_flag yes timer yes powman yes swcore yes psm yes then starts the power sequencer	RO	0x0

Bits	Description	Type	Reset
17	HAD_BOR: Last reset was from the brown-out detection block This resets: double_tap flag yes DP yes RPAP yes rescue_flag yes timer yes powman yes swcore yes psm yes then starts the power sequencer	RO	0x0
16	HAD_POR: Last reset was from the power-on reset This resets: double_tap flag yes DP yes RPAP yes rescue_flag yes timer yes powman yes swcore yes psm yes then starts the power sequencer	RO	0x0
15:5	Reserved.	-	-
4	RESCUE_FLAG: This is set by a rescue reset from the RP-AP. Its purpose is to halt before the bootrom before booting from flash in order to recover from a boot lock-up. The debugger can then attach once the bootrom has been halted and flash some working code that does not lock up.	WC	0x0
3:1	Reserved.	-	-
0	DOUBLE_TAP: This flag is set by double-tapping RUN. It tells bootcode to go into the bootloader.	RW	0x0

POWMAN: WDSEL Register

Offset: 0x30

Description

Allows a watchdog reset to reset the internal state of powman in addition to the power-on state machine (PSM).

Note that powman ignores watchdog resets that do not select at least the CLOCKS stage or earlier stages in the PSM. If using these bits, it's recommended to set PSM_WDSEL to all-ones in addition to the desired bits in this register. Failing to select CLOCKS or earlier will result in the POWMAN_WDSEL register having no effect.

Table 489. WDSEL Register

Bits	Description	Type	Reset
31:13	Reserved.	-	-
12	RESET_PSM: If set to 1, a watchdog reset will run the full power-on state machine (PSM) sequence From a user perspective it is the same as setting RSM_WDSEL_PROC_COLD From a hardware debug perspective it has the same effect as a reset from a glitch detector	RW	0x0
11:9	Reserved.	-	-

Bits	Description	Type	Reset
8	RESET_SWCORE: If set to 1, a watchdog reset will reset the switched core power domain and run the full power-on state machine (PSM) sequence From a user perspective it is the same as setting RSM_WDSEL_PROC_COLD From a hardware debug perspective it has the same effect as a power-on reset for the switched core power domain	RW	0x0
7:5	Reserved.	-	-
4	RESET_POWMAN: If set to 1, a watchdog reset will restore powman defaults, reset the timer, reset the switched core power domain and run the full power-on state machine (PSM) sequence This relies on clk_ref running. Use reset_powman_async if that may not be true	RW	0x0
3:1	Reserved.	-	-
0	RESET_POWMAN_ASYNC: If set to 1, a watchdog reset will restore powman defaults, reset the timer, reset the switched core domain and run the full power-on state machine (PSM) sequence This does not rely on clk_ref running	RW	0x0

POWMAN: SEQ_CFG Register

Offset: 0x34

Description

For configuration of the power sequencer

Writes are ignored while POWMAN_STATE_CHANGING=1

Table 490. SEQ_CFG Register

Bits	Description	Type	Reset
31:21	Reserved.	-	-
20	USING_FAST_POWCK: 0 indicates the POWMAN clock is running from the low power oscillator (32kHz) 1 indicates the POWMAN clock is running from the reference clock (2-50MHz)	RO	0x1
19:18	Reserved.	-	-
17	USING_BOD_LP: Indicates the brown-out detector (BOD) mode 0 = BOD high power mode which is the default 1 = BOD low power mode	RO	0x0
16	USING_VREG_LP: Indicates the voltage regulator (VREG) mode 0 = VREG high power mode which is the default 1 = VREG low power mode	RO	0x0
15:13	Reserved.	-	-
12	USE_FAST_POWCK: selects the reference clock (clk_ref) as the source of the POWMAN clock when switched-core is powered. The POWMAN clock always switches to the slow clock (lposc) when switched-core is powered down because the fast clock stops running. 0 always run the POWMAN clock from the slow clock (lposc) 1 run the POWMAN clock from the fast clock when available This setting takes effect when a power up sequence is next run	RW	0x1
11:9	Reserved.	-	-

Bits	Description	Type	Reset
8	RUN_LPOSC_IN_LP : Set to 0 to stop the low power osc when the switched-core is powered down, which is unwise if using it to clock the timer This setting takes effect when the swcore is next powered down	RW	0x1
7	USE_BOD_HP : Set to 0 to prevent automatic switching to bod high power mode when switched-core is powered up This setting takes effect when the swcore is next powered up	RW	0x1
6	USE_BOD_LP : Set to 0 to prevent automatic switching to bod low power mode when switched-core is powered down This setting takes effect when the swcore is next powered down	RW	0x1
5	USE_VREG_HP : Set to 0 to prevent automatic switching to vreg high power mode when switched-core is powered up This setting takes effect when the swcore is next powered up	RW	0x1
4	USE_VREG_LP : Set to 0 to prevent automatic switching to vreg low power mode when switched-core is powered down This setting takes effect when the swcore is next powered down	RW	0x1
3:2	Reserved.	-	-
1	HW_PWRUP_SRAM0 : Specifies the power state of SRAM0 when powering up swcore from a low power state (P1.xxx) to a high power state (P0.0xx). 0=power-up 1=no change	RW	0x0
0	HW_PWRUP_SRAM1 : Specifies the power state of SRAM1 when powering up swcore from a low power state (P1.xxx) to a high power state (P0.0xx). 0=power-up 1=no change	RW	0x0

POWMAN: STATE Register

Offset: 0x38

Description

This register controls the power state of the 4 power domains.

The current power state is indicated in POWMAN_STATE_CURRENT which is read-only.

To change the state, write to POWMAN_STATE_REQ.

The coding of POWMAN_STATE_CURRENT & POWMAN_STATE_REQ corresponds to the power states defined in the datasheet:

bit 3 = SWCORE

bit 2 = XIP cache

bit 1 = SRAM0

bit 0 = SRAM1

0 = powered up

1 = powered down

When POWMAN_STATE_REQ is written, the POWMAN_STATE_WAITING flag is set while the Power Manager determines what is required. If an invalid transition is requested the Power Manager will still register the request in POWMAN_STATE_REQ but will also set the POWMAN_BAD_REQ flag. It will then implement the power-up requests and ignore the power down requests. To do nothing would risk entering an unrecoverable lock-up state. Invalid requests are: any combination of power up and power down requests any request that results in swcore being powered and xip unpowered. If the request is to power down the switched-core domain then POWMAN_STATE_WAITING stays active until the processors halt. During this time the POWMAN_STATE_REQ field can be re-written to change or cancel the request. When the power state transition begins the POWMAN_STATE_WAITING_flag is cleared, the POWMAN_STATE_CHANGING flag is set and POWMAN register writes are ignored until the transition completes.

Table 491. STATE Register

Bits	Description	Type	Reset
31:14	Reserved.	-	-
13	CHANGING : Indicates a power state change is in progress	RO	0x0
12	WAITING : Indicates the power manager has received a state change request and is waiting for other actions to complete before executing it	RO	0x0
11	BAD_HW_REQ : Invalid hardware initiated state request, power up requests actioned, power down requests ignored	RO	0x0
10	BAD_SW_REQ : Invalid software initiated state request ignored	RO	0x0
9	PWRUP_WHILE_WAITING : Indicates that a power state change request was ignored because of a pending power state change request	WC	0x0
8	REQ_IGNORED : Indicates that a software state change request was ignored because it clashed with an ongoing hardware or debugger request	WC	0x0
7:4	REQ : This is written by software or hardware to request a new power state	RW	0x0
3:0	CURRENT : Indicates the current power state	RO	0xf

POWMAN: POW_FASTDIV Register

Offset: 0x3c

Table 492. POW_FASTDIV Register

Bits	Description	Type	Reset
31:11	Reserved.	-	-
10:0	divides the POWMAN clock to provide a tick for the delay module and state machines when clk_pow is running from the slow clock it is not divided when clk_pow is running from the fast clock it is divided by tick_div	RW	0x040

POWMAN: POW_DELAY Register

Offset: 0x40

Description

power state machine delays

Table 493. POW_DELAY Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:8	SRAM_STEP : timing between the sram0 and sram1 power state machine steps measured in units of the powman tick period ($\geq 1\mu\text{s}$), 0 gives a delay of 1 unit	RW	0x20
7:4	XIP_STEP : timing between the xip power state machine steps measured in units of the lpsc period, 0 gives a delay of 1 unit	RW	0x1
3:0	SWCORE_STEP : timing between the swcore power state machine steps measured in units of the lpsc period, 0 gives a delay of 1 unit	RW	0x1

POWMAN: EXT_CTRL0 Register

Offset: 0x44

Description

Configures a gpio as a power mode aware control output

Table 494. EXT_CTRL0 Register

Bits	Description	Type	Reset
31:15	Reserved.	-	-
14	LP_EXIT_STATE : output level when exiting the low power state	RW	0x0
13	LP_ENTRY_STATE : output level when entering the low power state	RW	0x0
12	INIT_STATE	RW	0x0
11:9	Reserved.	-	-
8	INIT	RW	0x0
7:6	Reserved.	-	-
5:0	GPIO_SELECT : selects from gpio 0→30 set to 31 to disable this feature	RW	0x3f

POWMAN: EXT_CTRL1 Register

Offset: 0x48

Description

Configures a gpio as a power mode aware control output

Table 495. EXT_CTRL1 Register

Bits	Description	Type	Reset
31:15	Reserved.	-	-
14	LP_EXIT_STATE : output level when exiting the low power state	RW	0x0
13	LP_ENTRY_STATE : output level when entering the low power state	RW	0x0
12	INIT_STATE	RW	0x0
11:9	Reserved.	-	-
8	INIT	RW	0x0
7:6	Reserved.	-	-
5:0	GPIO_SELECT : selects from gpio 0→30 set to 31 to disable this feature	RW	0x3f

POWMAN: EXT_TIME_REF Register

Offset: 0x4c

Description

Select a GPIO to use as a time reference, the source can be used to drive the low power clock at 32kHz, or to provide a 1ms tick to the timer, or provide a 1Hz tick to the timer. The tick selection is controlled by the POWMAN_TIMER register.

Table 496. EXT_TIME_REF Register

Bits	Description	Type	Reset
31:5	Reserved.	-	-
4	DRIVE_LPCK : Use the selected GPIO to drive the 32kHz low power clock, in place of LPOSC. This field must only be written when POWMAN_TIMER_RUN=0	RW	0x0
3:2	Reserved.	-	-

Bits	Description	Type	Reset
1:0	SOURCE_SEL : 0 → gpio12 1 → gpio20 2 → gpio14 3 → gpio22	RW	0x0
	Enumerated values:		
	0x0 → GPIO12		
	0x1 → GPIO20		
	0x2 → GPIO14		
	0x3 → GPIO22		

POWMAN: LPOSC_FREQ_KHZ_INT Register

Offset: 0x50

Description

Informs the AON Timer of the integer component of the clock frequency when running off the LPOSC.

Table 497.
LPOSC_FREQ_KHZ_INT
Register

Bits	Description	Type	Reset
31:6	Reserved.	-	-
5:0	Integer component of the LPOSC or GPIO clock source frequency in kHz. Default = 32 This field must only be written when POWMAN_TIMER_RUN=0 or POWMAN_TIMER_USING_XOSC=1	RW	0x20

POWMAN: LPOSC_FREQ_KHZ_FRAC Register

Offset: 0x54

Description

Informs the AON Timer of the fractional component of the clock frequency when running off the LPOSC.

Table 498.
LPOSC_FREQ_KHZ_FRAC
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	Fractional component of the LPOSC or GPIO clock source frequency in kHz. Default = 0.768 This field must only be written when POWMAN_TIMER_RUN=0 or POWMAN_TIMER_USING_XOSC=1	RW	0xc49c

POWMAN: XOSC_FREQ_KHZ_INT Register

Offset: 0x58

Description

Informs the AON Timer of the integer component of the clock frequency when running off the XOSC.

Table 499.
XOSC_FREQ_KHZ_INT
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	Integer component of the XOSC frequency in kHz. Default = 12000 Must be >1 This field must only be written when POWMAN_TIMER_RUN=0 or POWMAN_TIMER_USING_XOSC=0	RW	0x2ee0

POWMAN: XOSC_FREQ_KHZ_FRAC Register

Offset: 0x5c

Description

Informs the AON Timer of the fractional component of the clock frequency when running off the XOSC.

Table 500.
XOSC_FREQ_KHZ_FRA
C Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	Fractional component of the XOSC frequency in kHz. This field must only be written when POWMAN_TIMER_RUN=0 or POWMAN_TIMER_USING_XOSC=0	RW	0x0000

POWMAN: SET_TIME_63T048 Register

Offset: 0x60

Table 501.
SET_TIME_63T048
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	For setting the time, do not use for reading the time, use POWMAN_READ_TIME_UPPER and POWMAN_READ_TIME_LOWER. This field must only be written when POWMAN_TIMER_RUN=0	RW	0x0000

POWMAN: SET_TIME_47T032 Register

Offset: 0x64

Table 502.
SET_TIME_47T032
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	For setting the time, do not use for reading the time, use POWMAN_READ_TIME_UPPER and POWMAN_READ_TIME_LOWER. This field must only be written when POWMAN_TIMER_RUN=0	RW	0x0000

POWMAN: SET_TIME_31T016 Register

Offset: 0x68

Table 503.
SET_TIME_31T016
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	For setting the time, do not use for reading the time, use POWMAN_READ_TIME_UPPER and POWMAN_READ_TIME_LOWER. This field must only be written when POWMAN_TIMER_RUN=0	RW	0x0000

POWMAN: SET_TIME_15T00 Register

Offset: 0x6c

Table 504.
SET_TIME_15T00
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	For setting the time, do not use for reading the time, use POWMAN_READ_TIME_UPPER and POWMAN_READ_TIME_LOWER. This field must only be written when POWMAN_TIMER_RUN=0	RW	0x0000

POWMAN: READ_TIME_UPPER Register

Offset: 0x70

Table 505.
READ_TIME_UPPER
Register

Bits	Description	Type	Reset
31:0	For reading bits 63:32 of the timer. When reading all 64 bits it is possible for the LOWER count to rollover during the read. It is recommended to read UPPER, then LOWER, then re-read UPPER and, if it has changed, re-read LOWER.	RO	0x00000000

POWMAN: READ_TIME_LOWER Register

Offset: 0x74

Table 506.
READ_TIME_LOWER
Register

Bits	Description	Type	Reset
31:0	For reading bits 31:0 of the timer.	RO	0x00000000

POWMAN: ALARM_TIME_63T048 Register

Offset: 0x78

Table 507.
ALARM_TIME_63T048
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	This field must only be written when POWMAN_ALARM_ENAB=0	RW	0x0000

POWMAN: ALARM_TIME_47T032 Register

Offset: 0x7c

Table 508.
ALARM_TIME_47T032
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	This field must only be written when POWMAN_ALARM_ENAB=0	RW	0x0000

POWMAN: ALARM_TIME_31T016 Register

Offset: 0x80

Table 509.
ALARM_TIME_31T016
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	This field must only be written when POWMAN_ALARM_ENAB=0	RW	0x0000

POWMAN: ALARM_TIME_15T00 Register

Offset: 0x84

Table 510.
ALARM_TIME_15T00
Register

Bits	Description	Type	Reset
31:16	Reserved.	-	-
15:0	This field must only be written when POWMAN_ALARM_ENAB=0	RW	0x0000

POWMAN: TIMER Register

Offset: 0x88

Table 511. TIMER
Register

Bits	Description	Type	Reset
31:20	Reserved.	-	-
19	USING_GPIO_1HZ : Timer is synchronised to a 1hz gpio source	RO	0x0
18	USING_GPIO_1KHZ : Timer is running from a 1khz gpio source	RO	0x0
17	USING_LPOSC : Timer is running from lposc	RO	0x0
16	USING_XOSC : Timer is running from xosc	RO	0x0
15:14	Reserved.	-	-
13	USE_GPIO_1HZ : Selects the gpio source as the reference for the sec counter. The msec counter will continue to use the lposc or xosc reference.	RW	0x0
12:11	Reserved.	-	-
10	USE_GPIO_1KHZ : switch to gpio as the source of the 1kHz timer tick	SC	0x0
9	USE_XOSC : switch to xosc as the source of the 1kHz timer tick	SC	0x0
8	USE_LPOSC : Switch to lposc as the source of the 1kHz timer tick	SC	0x0
7	Reserved.	-	-
6	ALARM : Alarm has fired. Write to 1 to clear the alarm.	WC	0x0
5	PWRUP_ON_ALARM : Alarm wakes the chip from low power mode	RW	0x0
4	ALARM_ENAB : Enables the alarm. The alarm must be disabled while writing the alarm time.	RW	0x0
3	Reserved.	-	-
2	CLEAR : Clears the timer, does not disable the timer and does not affect the alarm. This control can be written at any time.	SC	0x0
1	RUN : Timer enable. Setting this bit causes the timer to begin counting up from its current value. Clearing this bit stops the timer from counting. Before enabling the timer, set the POWMAN_LPOSC_FREQ* and POWMAN_XOSC_FREQ* registers to configure the count rate, and initialise the current time by writing to SET_TIME_63T048 through SET_TIME_15T00. You must not write to the SET_TIME_x registers when the timer is running. Once configured, start the timer by setting POWMAN_TIMER_RUN=1. This will start the timer running from the LPOSC. When the XOSC is available switch the reference clock to XOSC then select it as the timer clock by setting POWMAN_TIMER_USE_XOSC=1	RW	0x0
0	NONSEC_WRITE : Control whether Non-secure software can write to the timer registers. All other registers are hardwired to be inaccessible to Non-secure.	RW	0x0

POWMAN: PWRUP0 Register

Offset: 0x8c**Description**

4 GPIO powerup events can be configured to wake the chip up from a low power state.

The pwrups are level/edge sensitive and can be set to trigger on a high/rising or low/falling event

The number of gpios available depends on the package option. An invalid selection will be ignored

source = 0 selects gpio0

1. +
2. + source = 47 selects gpio47
 source = 48 selects qspi_ss
 source = 49 selects qspi_sd0
 source = 50 selects qspi_sd1
 source = 51 selects qspi_sd2
 source = 52 selects qspi_sd3
 source = 53 selects qspi_sclk
 level = 0 triggers the pwrup when the source is low
 level = 1 triggers the pwrup when the source is high

Table 512. PWRUP0 Register

Bits	Description	Type	Reset
31:11	Reserved.	-	-
10	RAW_STATUS: Value of selected gpio pin (only if enable == 1)	RO	0x0
9	STATUS: Status of gpio wakeup. Write to 1 to clear a latched edge detect.	WC	0x0
8	MODE: Edge or level detect. Edge will detect a 0 to 1 transition (or 1 to 0 transition). Level will detect a 1 or 0. Both types of event get latched into the current_pwrup_req register.	RW	0x0
	Enumerated values:		
	0x0 → LEVEL		
	0x1 → EDGE		
7	DIRECTION	RW	0x0
	Enumerated values:		
	0x0 → LOW_FALLING		
	0x1 → HIGH_RISING		
6	ENABLE: Set to 1 to enable the wakeup source. Set to 0 to disable the wakeup source and clear a pending wakeup event. If using edge detect a latched edge needs to be cleared by writing 1 to the status register also.	RW	0x0
5:0	SOURCE	RW	0x3f

POWMAN: PWRUP1 Register**Offset:** 0x90**Description**

4 GPIO powerup events can be configured to wake the chip up from a low power state.

The pwrups are level/edge sensitive and can be set to trigger on a high/rising or low/falling event

The number of gpios available depends on the package option. An invalid selection will be ignored

source = 0 selects gpio0

1. +

2. + source = 47 selects gpio47
 source = 48 selects qspi_ss
 source = 49 selects qspi_sd0
 source = 50 selects qspi_sd1
 source = 51 selects qspi_sd2
 source = 52 selects qspi_sd3
 source = 53 selects qspi_sclk
 level = 0 triggers the pwrup when the source is low
 level = 1 triggers the pwrup when the source is high

Table 513. PWRUP1 Register

Bits	Description	Type	Reset
31:11	Reserved.	-	-
10	RAW_STATUS : Value of selected gpio pin (only if enable == 1)	RO	0x0
9	STATUS : Status of gpio wakeup. Write to 1 to clear a latched edge detect.	WC	0x0
8	MODE : Edge or level detect. Edge will detect a 0 to 1 transition (or 1 to 0 transition). Level will detect a 1 or 0. Both types of event get latched into the current_pwrup_req register.	RW	0x0
	Enumerated values:		
	0x0 → LEVEL		
	0x1 → EDGE		
7	DIRECTION	RW	0x0
	Enumerated values:		
	0x0 → LOW_FALLING		
	0x1 → HIGH_RISING		
6	ENABLE : Set to 1 to enable the wakeup source. Set to 0 to disable the wakeup source and clear a pending wakeup event. If using edge detect a latched edge needs to be cleared by writing 1 to the status register also.	RW	0x0
5:0	SOURCE	RW	0x3f

POWMAN: PWRUP2 Register

Offset: 0x94

Description

4 GPIO powerup events can be configured to wake the chip up from a low power state.
 The pwrups are level/edge sensitive and can be set to trigger on a high/rising or low/falling event
 The number of gpios available depends on the package option. An invalid selection will be ignored
 source = 0 selects gpio0

1. +
 2. + source = 47 selects gpio47
 source = 48 selects qspi_ss
 source = 49 selects qspi_sd0
 source = 50 selects qspi_sd1
 source = 51 selects qspi_sd2
 source = 52 selects qspi_sd3
 source = 53 selects qspi_sclk
 level = 0 triggers the pwrup when the source is low

level = 1 triggers the pwrup when the source is high

Table 514. PWRUP2 Register

Bits	Description	Type	Reset
31:11	Reserved.	-	-
10	RAW_STATUS : Value of selected gpio pin (only if enable == 1)	RO	0x0
9	STATUS : Status of gpio wakeup. Write to 1 to clear a latched edge detect.	WC	0x0
8	MODE : Edge or level detect. Edge will detect a 0 to 1 transition (or 1 to 0 transition). Level will detect a 1 or 0. Both types of event get latched into the current_pwrup_req register.	RW	0x0
	Enumerated values:		
	0x0 → LEVEL		
	0x1 → EDGE		
7	DIRECTION	RW	0x0
	Enumerated values:		
	0x0 → LOW_FALLING		
	0x1 → HIGH_RISING		
6	ENABLE : Set to 1 to enable the wakeup source. Set to 0 to disable the wakeup source and clear a pending wakeup event. If using edge detect a latched edge needs to be cleared by writing 1 to the status register also.	RW	0x0
5:0	SOURCE	RW	0x3f

POWMAN: PWRUP3 Register

Offset: 0x98

Description

4 GPIO powerup events can be configured to wake the chip up from a low power state.

The pwrups are level/edge sensitive and can be set to trigger on a high/rising or low/falling event

The number of gpios available depends on the package option. An invalid selection will be ignored

source = 0 selects gpio0

1. +
2. + source = 47 selects gpio47
 source = 48 selects qspi_ss
 source = 49 selects qspi_sd0
 source = 50 selects qspi_sd1
 source = 51 selects qspi_sd2
 source = 52 selects qspi_sd3
 source = 53 selects qspi_sclk
 level = 0 triggers the pwrup when the source is low
 level = 1 triggers the pwrup when the source is high

Table 515. PWRUP3 Register

Bits	Description	Type	Reset
31:11	Reserved.	-	-
10	RAW_STATUS : Value of selected gpio pin (only if enable == 1)	RO	0x0
9	STATUS : Status of gpio wakeup. Write to 1 to clear a latched edge detect.	WC	0x0

Bits	Description	Type	Reset
8	MODE: Edge or level detect. Edge will detect a 0 to 1 transition (or 1 to 0 transition). Level will detect a 1 or 0. Both types of event get latched into the current_pwrup_req register.	RW	0x0
	Enumerated values:		
	0x0 → LEVEL		
	0x1 → EDGE		
7	DIRECTION	RW	0x0
	Enumerated values:		
	0x0 → LOW_FALLING		
	0x1 → HIGH_RISING		
6	ENABLE: Set to 1 to enable the wakeup source. Set to 0 to disable the wakeup source and clear a pending wakeup event. If using edge detect a latched edge needs to be cleared by writing 1 to the status register also.	RW	0x0
5:0	SOURCE	RW	0x3f

POWMAN: CURRENT_PWRUP_REQ Register

Offset: 0x9c

Table 516.
CURRENT_PWRUP_REQ Register

Bits	Description	Type	Reset
31:7	Reserved.	-	-
6:0	Indicates current powerup request state pwrup events can be cleared by removing the enable from the pwrup register. The alarm pwrup req can be cleared by clearing timer.alarm_enab 0 = chip reset, for the source of the last reset see POWMAN_CHIP_RESET 1 = pwrup0 2 = pwrup1 3 = pwrup2 4 = pwrup3 5 = coresight_pwrup 6 = alarm_pwrup	RO	0x00

POWMAN: LAST_SWCORE_PWRUP Register

Offset: 0xa0

Table 517.
LAST_SWCORE_PWRUP
P Register

Bits	Description	Type	Reset
31:7	Reserved.	-	-
6:0	Indicates which pwrup source triggered the last switched-core power up 0 = chip reset, for the source of the last reset see POWMAN_CHIP_RESET 1 = pwrup0 2 = pwrup1 3 = pwrup2 4 = pwrup3 5 = coresight_pwrup 6 = alarm_pwrup	RO	0x00

POWMAN: DBG_PWRCFG Register

Offset: 0xa4

Table 518.
DBG_PWRCFG
Register

Bits	Description	Type	Reset
31:1	Reserved.	-	-
0	IGNORE: Ignore pwrup req from debugger. If pwrup req is asserted then this will prevent power down and set powerdown blocked. Set ignore to stop paying attention to pwrup_req	RW	0x0

POWMAN: BOOTDIS Register

Offset: 0xa8

Description

Tell the bootrom to ignore the BOOT0..3 registers following the next RSM reset (e.g. the next core power down/up).

If an early boot stage has soft-locked some OTP pages in order to protect their contents from later stages, there is a risk that Secure code running at a later stage can unlock the pages by powering the core up and down.

This register can be used to ensure that the bootloader runs as normal on the next power up, preventing Secure code at a later stage from accessing OTP in its unlocked state.

Should be used in conjunction with the OTP BOOTDIS register.

Table 519. BOOTDIS
Register

Bits	Description	Type	Reset
31:2	Reserved.	-	-
1	NEXT: This flag always ORs writes into its current contents. It can be set but not cleared by software. The BOOTDIS_NEXT bit is OR'd into the BOOTDIS_NOW bit when the core is powered down. Simultaneously, the BOOTDIS_NEXT bit is cleared. Setting this bit means that the BOOT0..3 registers will be ignored following the next reset of the RSM by powman. This flag should be set by an early boot stage that has soft-locked OTP pages, to prevent later stages from unlocking it by power cycling.	RW	0x0

Bits	Description	Type	Reset
0	NOW: When powman resets the RSM, the current value of BOOTDIS_NEXT is OR'd into BOOTDIS_NOW, and BOOTDIS_NEXT is cleared. The bootrom checks this flag before reading the BOOT0..3 registers. If it is set, the bootrom clears it, and ignores the BOOT registers. This prevents Secure software from diverting the boot path before a bootloader has had the chance to soft lock OTP pages containing sensitive data.	WC	0x0

POWMAN: DBGCONFIG Register

Offset: 0xac

Table 520.
DBGCONFIG Register

Bits	Description	Type	Reset
31:4	Reserved.	-	-
3:0	DP_INSTID: Configure DP instance ID for SWD multidrop selection. Recommend that this is NOT changed until you require debug access in multi-chip environment	RW	0x0

POWMAN: SCRATCH0, SCRATCH1, ..., SCRATCH6, SCRATCH7 Registers

Offsets: 0xb0, 0xb4, ..., 0xc8, 0xcc

Table 521. SCRATCH0,
SCRATCH1, ...,
SCRATCH6,
SCRATCH7 Registers

Bits	Description	Type	Reset
31:0	Scratch register. Information persists in low power mode	RW	0x00000000

POWMAN: BOOT0, BOOT1, BOOT2, BOOT3 Registers

Offsets: 0xd0, 0xd4, 0xd8, 0xdc

Table 522. BOOT0,
BOOT1, BOOT2,
BOOT3 Registers

Bits	Description	Type	Reset
31:0	Scratch register. Information persists in low power mode	RW	0x00000000

POWMAN: INTR Register

Offset: 0xe0

Description

Raw Interrupts

Table 523. INTR
Register

Bits	Description	Type	Reset
31:4	Reserved.	-	-
3	PWRUP_WHILE_WAITING: Source is state.pwrup_while_waiting	RO	0x0
2	STATE_REQ_IGNORED: Source is state.req_ignored	RO	0x0
1	TIMER	RO	0x0
0	VREG_OUTPUT_LOW	WC	0x0

POWMAN: INTE Register

Offset: 0xe4

Description

Interrupt Enable

Table 524. INTE Register

Bits	Description	Type	Reset
31:4	Reserved.	-	-
3	PWRUP_WHILE_WAITING: Source is state.pwrup_while_waiting	RW	0x0
2	STATE_REQ_IGNORED: Source is state.req_ignored	RW	0x0
1	TIMER	RW	0x0
0	VREG_OUTPUT_LOW	RW	0x0

POWMAN: INTF Register

Offset: 0xe8

Description

Interrupt Force

Table 525. INTF Register

Bits	Description	Type	Reset
31:4	Reserved.	-	-
3	PWRUP_WHILE_WAITING: Source is state.pwrup_while_waiting	RW	0x0
2	STATE_REQ_IGNORED: Source is state.req_ignored	RW	0x0
1	TIMER	RW	0x0
0	VREG_OUTPUT_LOW	RW	0x0

POWMAN: INTS Register

Offset: 0xec

Description

Interrupt status after masking & forcing

Table 526. INTS Register

Bits	Description	Type	Reset
31:4	Reserved.	-	-
3	PWRUP_WHILE_WAITING: Source is state.pwrup_while_waiting	RO	0x0
2	STATE_REQ_IGNORED: Source is state.req_ignored	RO	0x0
1	TIMER	RO	0x0
0	VREG_OUTPUT_LOW	RO	0x0

6.5. Power Reduction Strategies

RP2350 retains the SLEEP and DORMANT states for dynamic power control from RP2040. It extends these states by introducing power domains (Section 6.2.1), which allow power to be removed from various components on chip, virtually eliminating the leakage currents, and allowing lower power modes to be supported.

6.5.1. Top-level Clock Gates

Each clock domain (for example, the system clock) may drive a large number of distinct hardware blocks, not all of which may be required at once. To avoid unnecessary power dissipation, each individual endpoint of each clock (for example, the UART system clock input) may be disabled at any time.

Enabling and disabling a clock gate is glitch-free. If a peripheral clock is temporarily disabled, and subsequently re-enabled, the peripheral will be in the same state as prior to the clock being disabled. No reset or reinitialisation should be required.

Clock gates are controlled by two sets of registers: the **WAKE_ENx** registers (starting at **WAKE_EN0**) and **SLEEP_ENx** registers (starting at **SLEEP_EN0**). These two sets of registers are identical at the bit level, each possessing a flag to control each clock endpoint. The **WAKE_EN** registers specify which clocks are enabled whilst the system is awake, and the **SLEEP_ENx** registers select which clocks are enabled while the processor is in the SLEEP state (Section 6.5.2).

The two processors do not have externally-controllable clock gates. Instead, the processors gate the clocks of their subsystems autonomously, based on execution of **WFI/WFE** instructions, and external Event and IRQ signals.

6.5.2. SLEEP State

RP2350 enters the SLEEP state when all of the following are true:

- Both processors are asleep (e.g. in a **WFE** or **WFI** instruction)
- The system DMA has no outstanding transfers on any channel

RP2350 exits the SLEEP state when either processor is awoken by an interrupt.

When in the SLEEP state, the top-level clock gates are masked by the **SLEEP_ENx** registers (starting at **SLEEP_EN0**), rather than the **WAKE_ENx** registers (starting at **WAKE_EN0**). This permits more aggressive pruning of the clock tree when the processors are asleep.

i NOTE

Though it is possible for a clock to be enabled during SLEEP and disabled outside of SLEEP, this is generally not useful.

For example, if the system is sleeping until a character interrupt from a UART, the entire system except for the UART can be clock-gated (**SLEEP_ENx** = all-zeroes except for **CLK_SYS_UART0** and **CLK_PERI_UART0**). This includes system infrastructure such as the bus fabric.

When the UART asserts its interrupt and wakes a processor, RP2350 leaves SLEEP mode and switches back to the **WAKE_ENx** clock mask. At the minimum, this should include the bus fabric and the memory devices containing the processor's stack and interrupt vectors.

A system-level clock request handshake holds the processors off the bus until the clocks are re-enabled.

6.5.3. DORMANT State

The DORMANT state is a true zero-dynamic-power sleep state, where all clocks (and all oscillators) are disabled. The system can awake from the DORMANT state upon a GPIO event (high/low level or rising/falling edge), or an AON Timer alarm: this restarts one of the oscillators (either ring oscillator or crystal oscillator) and ungates the oscillator output once it is stable. System state is retained, so code execution resumes immediately upon leaving the DORMANT state.

If relying on the AON Timer (Section 12.10) to wake from the DORMANT state, the AON Timer must run from the LPOSC or an external clock source. The AON Timer accepts clock frequencies as low as 1Hz.

DORMANT does not halt PLLs. To avoid unnecessary power dissipation, software should power down PLLs before entering the DORMANT state, and power up and reconfigure the PLLs again after exiting.

If you halt the crystal oscillator (XOSC), you must also halt the PLLs to prevent them losing lock when their input reference clock stops. The PLL VCO may behave erratically when the frequency reference is lost, such as increasing to a very high frequency. Reconfigure and re-enable the PLLs after the XOSC starts again. Do not attempt to run clocks from the PLLs while the XOSC is stopped.

The DORMANT state is entered by writing a keyword to the DORMANT register in whichever oscillator is active: ring oscillator (Section 8.3) or crystal oscillator (Section 8.2). If both are active, the one providing the processor clock must be stopped last because it will stop software from executing.

6.5.3.1. Waking from the DORMANT State

The system exits the DORMANT state on any of the following events:

- an alarm from the AON Timer which causes `TIMER.ALARM` to assert
- the assertion of an interrupt from GPIO Bank 0 to the `DORMANT_WAKE` interrupt destination
- the assertion of an interrupt from GPIO Bank 1 to the `DORMANT_WAKE` interrupt destination

When waking from the AON Timer you do not have to enable the IRQ output from POWMAN. It is sufficient for the timer to fire, without being mapped to an interrupt output. Any AON Timer alarm comparison event which causes `TIMER.ALARM` to assert causes the system to exit the DORMANT state. It is the actual alarm event which causes the exit, not the `TIMER.ALARM` status; if you enter the DORMANT state with the `TIMER.ALARM` status set to 1, but the timer alarm comparison logic *disabled* by `TIMER.ALARM_ENAB`, you will not exit the DORMANT state.

The GPIO Bank registers have interrupt enable registers for interrupts targeting the DORMANT mode wake logic, such as `DORMANT_WAKE_INTE0`. These are identical to the interrupt enable registers for interrupts targeting the processors, such as `PROCO_INTE0`.

Waking from the DORMANT state restarts the oscillator which was disabled by entry to the DORMANT state. It does not restart any other oscillators, or change any system-level clock configuration.

6.5.4. Memory Periphery Power Down

The main system memories (`SRAM0` → `SRAM9`, mapped to bus addresses `0x20000000` to `0x20081fff`), as well as the USB DPRAM, can be partially powered down via the `MEMPOWERDOWN` register in the SYSCFG registers (see Section 12.15.2). This powers down the analogue circuitry used to access the SRAM storage array (the **periphery** of the SRAM) but the storage array itself remains powered. Memories retain their current contents, but cannot be accessed. Static power is reduced.

CAUTION

Memories must not be accessed when powered down. Doing so can corrupt memory contents.

When powering a memory back up, a 20ns delay is required before accessing the memory again.

The XIP cache (see Section 4.4) can also be powered down, with `CTRL.POWER_DOWN`. The XIP hardware will not generate cache accesses whilst the cache is powered down. Note that this is unlikely to produce a net power savings if code continues to execute from XIP, due to the comparatively high voltages and switching capacitances of the external QSPI bus.

6.5.5. Full Memory Power Down

RP2350 can completely power down its internal SRAM. Unlike the memory periphery power down described in Section 6.5.4, this completely disconnects the SRAM from the power supply, reducing static power to near zero.

Contents are lost when fully powering down memories. When you power memories up again following a power down,

the contents is completely undefined.

There are three distinct SRAM power domains:

SRAM0

Contains main system SRAM for addresses `0x20000000` through `0x2003ffff` (SRAM banks 0 through 3).

SRAM1

Contains main system SRAM for addresses `0x20040000` through `0x20081fff` (SRAM banks 4 through 9).

XIP

Contains the XIP cache and the boot RAM.

The XIP power domain is always powered when the switched core domain is powered. The switched core domain is the domain which includes all core logic, such as processors, bus fabric and peripherals. This means the memories in this domain are always powered whenever software is running.

Besides powering memory down to save power, you can also leave memories powered *up* whilst powering down the switched core domain. This retains program state in SRAM while eliminating static power dissipation in core logic.

For more information see:

- [Chapter 4](#) for a list of RP2350 memory resources, including main system SRAM, the XIP cache and boot RAM
- [Section 6.2.1](#) for the definition of core power domains, including the memory power domains enumerated above
- [Section 6.2.2](#) for the list of supported memory power states
- [Section 6.2.3](#) for information on initiating power state transitions to power memories up or down
- [Section 14.9.7.2](#) for typical power consumption in low-power states including memory power down

6.5.6. Programmer's Model

6.5.6.1. Sleep

The `hello_sleep` example ([hello_sleep_aon.c in the pico-playground GitHub repository](#)) demonstrates sleep mode. The `hello_sleep` application (and underlying functions) takes the following steps:

1. Switches all clocks in the system to run from XOSC.
2. Configures an alarm in the AON Timer for 10 seconds in the future.
3. Sets the AON Timer clock as the only clock running in sleep mode using the `SLEEP_ENx` registers (see `SLEEP_EN0`).
4. Enables deep sleep in the processor.
5. Calls `__wfi` on processor, which will put the processor into deep sleep until woken by the AON Timer interrupt.
6. After 10 seconds, the AON Timer interrupt clears the alarm and then calls a user supplied callback function.
7. The callback function ends the example application.

i NOTE

To enter sleep mode, you must enable deep sleep on both `proc0` and `proc1`, call `__wfi`, and ensure the DMA is stopped.

`hello_sleep` makes use of functions in `pico_sleep` of the [Pico Extras](#). In particular, `sleep_goto_sleep_until` puts the processor to sleep until woken up by an AON Timer time assumed to be in the future.

Pico Extras: https://github.com/raspberrypi/pico-extras/blob/master/src/rp2_common/pico_sleep/sleep.c Lines 159 - 183

```

159 void sleep_goto_sleep_until(struct timespec *ts, aon_timer_alarm_handler_t callback)
160 {
161
162     // We should have already called the sleep_run_from_dormant_source function
163     // This is only needed for dormancy although it saves power running from xosc while
    sleeping
164     //assert(dormant_source_valid(_dormant_source));
165
166     clocks_hw->sleep_en0 = CLOCKS_SLEEP_EN0_CLK_REF_POWMAN_BITS;
167     clocks_hw->sleep_en1 = 0x0;
168
169     aon_timer_enable_alarm(ts, callback, false);
170
171     stdio_flush();
172
173     // Enable deep sleep at the proc
174     processor_deep_sleep();
175
176     // Go to sleep
177     __wfi();
178 }

```

6.5.6.2. DORMANT

The `hello_dormant` example, `hello_dormant_gpio.c` in the [pico-playground GitHub repository](#), demonstrates the DORMANT state. The example takes the following steps:

1. Switches all clocks in the system to run from XOSC.
2. Configures a GPIO interrupt for the `dormant_wake` hardware, which can wake both the ROSC and XOSC from dormant mode.
3. Puts the XOSC into dormant mode, which stops all processor execution (and all other clocked logic on the chip) immediately.
4. When GPIO 10 goes high, the XOSC restarts and program execution continues.

`hello_dormant` uses `sleep_goto_dormant_until_pin` under the hood:

Pico Extras: https://github.com/raspberrypi/pico-extras/blob/master/src/rp2_common/pico_sleep/sleep.c Lines 258 - 282

```

258 void sleep_goto_dormant_until_pin(uint gpio_pin, bool edge, bool high) {
259     bool low = !high;
260     bool level = !edge;
261
262     // Configure the appropriate IRQ at IO bank 0
263     assert(gpio_pin < NUM_BANK0_GPIOS);
264
265     uint32_t event = 0;
266
267     if (level && low) event = IO_BANK0_DORMANT_WAKE_INTE0_GPIO0_LEVEL_LOW_BITS;
268     if (level && high) event = IO_BANK0_DORMANT_WAKE_INTE0_GPIO0_LEVEL_HIGH_BITS;
269     if (edge && high) event = IO_BANK0_DORMANT_WAKE_INTE0_GPIO0_EDGE_HIGH_BITS;
270     if (edge && low) event = IO_BANK0_DORMANT_WAKE_INTE0_GPIO0_EDGE_LOW_BITS;
271
272     gpio_init(gpio_pin);
273     gpio_set_input_enabled(gpio_pin, true);
274     gpio_set_dormant_irq_enabled(gpio_pin, event, true);
275 }

```

```
276     _go_dormant();  
277     // Execution stops here until woken up  
278  
279     // Clear the irq so we can go back to dormant mode again if we want  
280     gpio_acknowledge_irq(gpio_pin, event);  
281     gpio_set_input_enabled(gpio_pin, false);  
282 }
```